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**Transmission Line Circuit for Simplified Hodgkin-Huxley
Neuron Model**

By

Kruthika Yogeesh Gowda

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Supervisor: **Dr. Harold Chong**

Second Examiner: **Dr. Rosalia Moreddu**

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Abstract

This dissertation presents the design and simulation of an analog neuron model that replicates the functional stages of dendrites, soma, and axon using continuous-time RC and operational amplifier circuits. The objective of the project was to investigate how biologically inspired filtering, spiking, and transmission behaviours can be reproduced in hardware, and to evaluate both the qualitative features and quantitative performance of the model. Simulation work was completed using Multisim for circuit-level design and MATLAB for frequency and time-domain analysis.

To validate the performance of the model, four biologically relevant input signals were applied: Hodgkin–Huxley waveforms, EEG signals, PQRS sequences, and multi-sinusoidal test signals. The dendritic circuits demonstrated selective band-pass filtering in the alpha frequency range between 9 Hz and 13 Hz. At the soma, spike initiation was achieved with an average width of 4.07 ms, consistent with the biological range of 2–5 ms. Quantitative analysis showed that soma-to-axon signal propagation introduced a mean SNR loss of only 0.35 dB, while the overall SNR at the input stage was –6.86 dB. Root-mean-square error was measured at 0.021 V and jitter across the soma–axon interface was approximately 0.94 ms. These results confirm that the circuit maintains strong fidelity in spike generation and transmission, with the largest distortions confined to the dendritic filtering stage.

The study further compared design choices to identify optimal operating conditions. The results indicate that frequency preservation and timing precision are best achieved when dendritic filter parameters are tuned for alpha-band selectivity, and when comparator thresholds are scaled appropriately to emulate biological firing conditions. Overall, the project demonstrates that essential neuronal behaviours can be reproduced with simple RC and op-amp circuits, providing a reproducible framework for analog neuromorphic modelling and suggesting future applications in low-power neuromorphic devices.

Keywords: Analog neuron model, dendritic filtering, spike generation, jitter, SNR, Multisim, MATLAB, neuromorphic circuits

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Chapter 1: Introduction

1.1 Background

The human brain is an exceptionally efficient computational system, capable of performing perception, decision-making, and control with very low energy requirements. Its functionality arises from interconnected neurons that receive signals through dendrites, process them at the soma, and transmit output through axons. These biological processes inspire the field of neuromorphic engineering, which aims to reproduce neural computation using hardware systems [1], [2].

Mathematical models such as the Hodgkin–Huxley equations describe ion channel dynamics and membrane potentials with high accuracy [6]. While these models provide biological realism, they are mathematically intensive and unsuitable for direct hardware implementation. Digital simulations are more practical but remain resource-intensive when scaled to large networks. In contrast, analog hardware models offer real-time operation, continuous signal behaviour, and reduced power consumption, making them suitable for exploring simplified representations of neural activity [2], [18].

By building circuit equivalents of dendrites, soma, and axons, it is possible to investigate fundamental input–output behaviours of neurons within electronic systems. Previous studies have demonstrated that tuned RC band-pass filters can replicate dendritic frequency selectivity [1], comparator circuits can mimic threshold-based spiking [6], and RC ladder networks with periodic amplification can represent axonal transmission [19], [20]. These findings provide a foundation for circuit-level neuron models that balance simplicity with functional realism.

1.2 Motivation and Rationale

The motivation for this project arises from the need to design simplified yet biologically informed hardware models of neurons that can be implemented using standard electronic components. Traditional mathematical and digital models such as Hodgkin–Huxley describe neural activity with high fidelity but are computationally intensive and unsuitable for real-time hardware use [6]. Many neuromorphic systems also rely on complex or resource-heavy architectures, limiting their accessibility for wider experimentation [2], [9].

This dissertation makes a contribution by demonstrating how essential neuronal behaviours can be reproduced in a modular analog framework. By dividing the model into dendritic filtering, soma spiking, and axonal transmission, the project shows how circuit-level components can approximate measurable benchmarks such as spike width, alpha-band selectivity, and soma–axon jitter. The immediate value lies in providing a hardware-realistic approach that highlights both the capabilities and the limitations of analog neuron circuits.

The longer-term motivation is to provide groundwork for the development of energy-efficient and biologically inspired computing systems. While this project does not aim to replicate full ionic channel dynamics or advanced synaptic mechanisms, it demonstrates how circuit-level constraints influence neural realism. Such insights may inform future neuromorphic hardware design, contribute to teaching and research on simplified neuron models, and open pathways for accessible hardware demonstrations that complement theoretical neuroscience [9], [21].

1.3 Scope and Objectives

This dissertation focuses on the design and validation of an analog neuron circuit using Multisim for simulation and MATLAB for analysis. The scope of the work is limited to three main modules:

- **Dendritic filtering:** implemented with tuned RC band-pass circuits designed to pass signals within the alpha frequency range (8–13 Hz) while attenuating higher and lower frequencies.
- **Soma spiking:** achieved with comparator and Schmitt trigger circuits that generate reproducible digital spikes when a defined threshold is crossed.
- **Axonal transmission:** represented by an RC ladder with periodic amplification to maintain spike timing and shape during propagation.

The project does not attempt to replicate the full Hodgkin–Huxley model or include synaptic plasticity mechanisms. Instead, it focuses on reproducing measurable neuronal behaviours that demonstrate the feasibility of circuit-based analog neurons.

The objectives of the project are:

1. To design circuit equivalents of dendritic filters, soma spiking stages, and axonal transmission pathways using standard components.
2. To simulate the frequency response, spike initiation, and transmission behaviour of each module.
3. To compare circuit performance with biological benchmarks such as spike width, dendritic passband, and soma-axon jitter.
4. To evaluate how circuit constraints affect biological similarity and to identify areas for improvement in future designs.

Chapter 2: Literature Review

This chapter provides a comprehensive review of the biological principles, engineering strategies, and prior research that inform the design of the proposed analog neuron model. Beginning with the structural and functional properties of biological neurons, the review then explores how these behaviours can be translated into circuit components such as RC filters, comparators, and Schmitt triggers. Existing models across analog, digital, and memristive domains are critically examined to highlight their strengths and limitations. The chapter concludes with a focused gap analysis that justifies the current project's architectural and methodological choices in addressing biological realism, signal fidelity, and implementation feasibility.

2.1 Biological Neuron Structure and Signal Pathway

Neurons are the foundational units of the nervous system, responsible for processing and transmitting information via electrical and chemical signals. Structurally, a neuron comprises dendrites, a soma (cell body), and an axon. Each substructure plays a distinct and critical role in the neural signal pathway. Dendrites receive synaptic inputs and are responsible for signal integration and filtering. The soma functions as the processing core where action potentials are initiated once the membrane potential exceeds a threshold. The axon propagates these action potentials toward downstream neurons, ensuring that the integrity of the signal is preserved over long distances [6].

In biological systems, dendrites can span large distances and host thousands of synapses, each contributing to the neuron's response through spatial and temporal summation. Temporal summation occurs when multiple postsynaptic potentials arrive in quick succession at the same synapse, while spatial summation integrates signals from multiple synaptic locations. This integration of synaptic inputs results in complex voltage dynamics, modulated by passive and active dendritic properties [6]. The filtering that occurs in dendrites acts as a pre-processing mechanism, shaping the frequency spectrum of the input signals before they reach the soma. This biological filtering is essential to remove transient noise and prevent the neuron from responding to insignificant fluctuations.

The soma integrates these filtered inputs and determines whether the neuron will generate an output spike. If the combined input is sufficient to push the membrane potential beyond a critical threshold, an all-or-none spike is generated. This mechanism is driven by the opening of voltage-gated sodium channels, leading to rapid depolarization and a cascading action potential. The axon then serves as a transmission channel for the action potential, actively regenerating the signal at intervals to prevent attenuation over long distances. This regeneration is achieved through the presence of Nodes of Ranvier, where action potentials are re-amplified [6].

Figure 2.1 shows a labelled anatomical diagram of a biological neuron, depicting the key structural components such as dendrites, soma, axon, and Nodes of Ranvier.

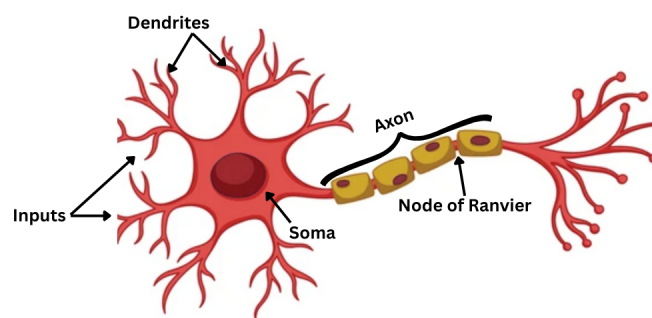


Figure 2.1: Basic structure of a biological neuron showing dendrites, soma, axon, and nodes of Ranvier.

A hardware model inspired by biology must preserve three defining behaviours: dendritic filtering, threshold-triggered spiking, and proper axonal propagation. These properties not only support the functional requirements of neural signalling but also serve as benchmarks for validating any hardware realization of a neuron. In practical circuit designs, the system must also demonstrate robustness to non-ideal conditions such as noise, signal overlap, and input variability. These factors reflect the unpredictable nature of biological environments and real-world applications. Therefore, the analog circuit must accommodate a range of inputs, such as EEG signals or synthetic multi-frequency patterns, without distortion or instability [7], [8].

2.2 Dendritic Filtering and RC Modelling

Dendritic processing is more than passive conduction. It attenuates high-frequency inputs while integrating lower-frequency signals, thereby emphasizing biologically relevant timing and input diversity. This behaviour can be modelled using first-order RC circuits, where resistance simulates leaky ion channels and capacitance models membrane charge storage [1]. RC circuits behave as low-pass filters that suppress high-frequency components while allowing slower signals to pass. This mirrors the biological tendency to ignore fast, noisy transients in favour of sustained inputs.

The time constant $\tau = RC$ is central to this behaviour. It determines the cut-off frequency below which signals are transmitted with minimal loss. By tuning R and C, the circuit can be made to allow specific frequency bands such as the alpha (8 - 12 Hz) or beta (13 - 30 Hz) ranges seen in signals, while attenuating noise and irrelevant oscillations [7]. A time constant of 0.1 to 0.2 seconds would typically correspond to low-pass filtering in the alpha range. These frequency characteristics can be experimentally verified using Bode plots in simulation tools, allowing precise control over signal fidelity.

In actual dendritic trees, filtering varies across branches due to differences in geometry and ion channel density. Analog circuits can replicate this by using cascaded or parallel RC stages, each with tailored time constants. This approach supports selective integration, where different branches simulate local versus global summation. A longer time constant in one branch mimics slower temporal integration, while a shorter time constant simulates rapid local responses [26].

Advanced analog implementations include nonlinear dendritic responses using transconductance amplifiers or memristive devices. However, for core filtering functionality, simple RC configurations remain sufficient. They offer stability, ease of tuning, and reproducibility across various biological signal types [18], [22]. By using a modular RC stage for each dendritic input in this project, frequency filtering is biologically mapped and mathematically grounded. Signals entering the soma stage have already undergone noise rejection, and only meaningful temporal components are preserved.

This also makes the design extensible. Additional branches can be added with different RC values to test responses across multiple frequency bands. This approach captures the diversity seen in real neurons and enhances the model's adaptability for future signal types or learning mechanisms.

2.3 Axonal Spiking: Threshold Dynamics and Signal Transmission

Neurons rely on precise threshold-triggered spiking to ensure meaningful signal propagation. In biological systems, this is governed by voltage-gated ion channels. In analog hardware, this behaviour is effectively replicated using comparators and Schmitt triggers. Comparators output a high voltage when input exceeds a set threshold, emulating the soma's decision-making. Schmitt triggers further enhance stability through hysteresis, which prevents noise-induced flickering near threshold values [4], [12], [13].

When the input crosses the upper threshold, the circuit switches to the high state and remains there until the input drops below the lower threshold. This mechanism mimics the refractory period of neurons, ensuring only well-defined spikes are produced and eliminating false positives caused by input jitter. The threshold values in Schmitt triggers can be set to mimic biological firing rates, with proper tuning based on the desired response time.

Once a spike is generated, it must have consistent width and amplitude to remain interpretable by downstream systems. Monostable multivibrators or one-shot pulse generators accomplish this by shaping each spike into a uniform pulse, independent of input duration [27]. The pulse width is defined by RC timing or feedback parameters, ensuring repeatability. This feature is essential for creating spike trains that preserve the temporal characteristics of the original input.

To simulate the role of the axon and Nodes of Ranvier, op-amps are used in non-inverting amplifier configurations. These stages restore signal strength and isolate outputs, maintaining spike integrity throughout the propagation path [19]. Amplified output stages also ensure compatibility with future downstream blocks, such as summation nodes or synaptic weighting units. In simulation environments, this setup guarantees that signal degradation is avoided even after multiple propagation stages.

The axon's output waveform is expected to be clean, noise-free, and pulse-locked to input thresholds. The design must also support dynamic inputs, including high-frequency bursts, overlapping waveforms, or non-periodic activity. The model used here has been tuned in Multisim to produce spike trains consistent with biological behaviour across these diverse conditions.

2.4 Analog Mapping of Neuronal Components

A key aspect of implementing bio-inspired neuron models in hardware lies in the accurate abstraction of biological elements into circuit equivalents. The electrical behaviour of a neuron's membrane potential is analogized to voltage, ionic current to current flow, and ion channel gating to threshold-based circuit operations. In this analog domain, time constants of neural membranes directly correspond to RC circuits, enabling the designer to simulate integrative behaviour without software or microcontroller intervention [26], [28].

The significance of this mapping lies in the biological authenticity it enables. The dendrite, modelled as an RC low-pass filter, contributes frequency selectivity and noise suppression. The soma is mimicked using a comparator with precise threshold values and hysteresis, effectively translating continuous graded potentials into discrete spike outputs. The axon is realized through an amplifier stage and monostable pulse shaping circuit, facilitating uniform spike propagation that resembles axonal regeneration seen in Nodes of Ranvier [6], [19].

A major design consideration in this analog mapping is maintaining waveform continuity. While digital systems rely on sampling and discretization, analog systems preserve signal shape and temporal alignment across all circuit stages. This becomes especially important in applications requiring frequency domain fidelity, such as EEG decoding or biofeedback systems. In the presented model, every spike is the result of a real-time decision made by the soma stage, processed through biologically grounded filters and regenerated with defined amplitude and timing constraints.

Furthermore, analog models offer scalability advantages. Unlike digital processors that require increased computational power for complex dynamics, analog systems scale in hardware blocks. This means additional dendritic branches or threshold stages can be incorporated with minimal changes to the overall architecture. Such modularity makes analog neuron models suitable for neuromorphic arrays and circuit-level learning experiments where individual neurons can be tuned independently.

By preserving the hierarchical structure of biological signalling and mimicking the shape, timing, and propagation of spikes, analog mapping provides a biologically faithful yet hardware-accessible neuron model that can be replicated across different platforms or test setups.

2.5 Existing Neuron Models

Numerous circuit models have been proposed to emulate the behaviour of biological neurons. These models broadly fall into three categories: digital, analog, and memristor-based.

Digital neuron models utilise programmable microcontrollers or FPGAs to simulate neural dynamics. These are highly flexible and support large-scale networks with programmable learning rules and plasticity mechanisms [10]. They are widely used in neuromorphic platforms such as IBM TrueNorth and Intel Loihi due to their scalability. However, they operate in discrete time steps and rely on clocked execution, introducing latency and limiting real-time biological mimicry.

Analog neuron models attempt to capture continuous-time neuronal dynamics using standard electronic components. Notable implementations include silicon neurons built from transistor-capacitor networks that model ion channel activity [13], [26]. These systems provide high biological fidelity, particularly in simulating subthreshold membrane dynamics and action potential generation. However, their reliance on precision tuning, component density, and intricate layout design makes them difficult to implement in accessible or resource-constrained settings.

Memristor-based models are inspired by the resistive switching and memory-retaining characteristics of memristive devices. They allow for compact designs and can support both synaptic and neuronal functions [3], [5], [10]. Memristive circuits are highly attractive for long-term state retention and spike-timing-dependent plasticity. Despite their potential, the technology is still emerging and not yet widely supported by standard fabrication flows or circuit simulators. Device variability, nonlinearity, and lack of commercial accessibility limit their widespread adoption.

Each of these neuron model types offers specific strengths, whether it be scalability, biological realism, or non-volatility. However, none singularly balances all design constraints such as reproducibility, simplicity, accessibility, and frequency-specific biological behaviour. This necessitates the development of an alternative approach that incorporates essential biological properties while remaining feasible for analog implementation.

2.6 Gap Analysis and Justification of Present Work

The review of existing neuron models demonstrates considerable progress in replicating various aspects of biological function. However, several limitations remain that directly motivate the present work.

A first limitation is the lack of frequency-domain processing in many models. Real neurons exhibit selective responses to input frequency bands [1], [8]. Yet most digital and hybrid models prioritise flexibility and network integration over frequency-selective filtering [15], [16]. This restricts their suitability for applications involving biological signals such as EEG, where band-specific dynamics are critical [7].

A second limitation concerns waveform fidelity. In many digital and analog designs, spikes are represented as binary events, with emphasis placed on timing while width and amplitude are neglected [12], [23]. Biological neurons, by contrast, produce spikes with consistent shape and width, and deviations affect downstream processing [6]. Few existing models explicitly validate benchmarks such as jitter or spike width preservation, with most memristive circuits instead focusing on spiking diversity or plasticity [3], [5], [17].

A third limitation relates to modularity and scalability. Analog silicon neurons [13], [26] and memristive systems [19], [20] often adopt monolithic designs that are difficult to extend or adapt for new signal types. In contrast, modular architectures, where dendrites, soma, and axon are represented independently, provide flexibility for both experimentation and replication. Such modularity is particularly valuable in analog systems, where additional branches can simulate the diversity of dendritic or axonal pathways [25].

Finally, accessibility and reproducibility remain major challenges. Memristor-based neurons require specialised fabrication processes and suffer from device variability [22]. CMOS-based designs achieve scalability but at the cost of high power and custom layouts [28], [29]. Digital processors, although efficient, do not preserve continuous-time dynamics [10]. There is therefore a need for neuron models that use only standard components, are implementable in widely available simulation environments, and preserve biologically relevant dynamics.

The present project addresses these gaps by introducing a modular analog neuron constructed entirely from resistors, capacitors, comparators, and operational amplifiers. Dendritic RC filters provide frequency-selective input conditioning [26], comparators with hysteresis generate consistent threshold-triggered spikes [18], and RC ladders with amplification reproduce axonal regeneration [20], [23]. The model is validated against measurable benchmarks including dendritic passband selectivity, spike width, jitter, and signal-to-noise ratio, demonstrating that biologically relevant transmission fidelity can be achieved in hardware.

By focusing specifically on axonal transmission, the project highlights an area that has received comparatively little attention in neuromorphic literature. Previous work has concentrated on spiking or synaptic behaviour [3], [4], [5], [17], while the fidelity of spike propagation across axons has rarely been treated as a primary design concern [19], [20], [23]. This dissertation makes a distinct contribution by showing that transmission fidelity can be realised using simple, reproducible analog circuits, thereby establishing a foundation for future neuromorphic models that prioritise both accessibility and biological realism.

Chapter 3: Theoretical Modelling

The objective of this chapter is to provide a biologically grounded yet circuit-realistic theoretical model of the neuron system. Drawing inspiration from neurophysiology and established electrophysiological models, this chapter formulates the analog equivalents of the dendritic tree, soma thresholding, axonal conduction, and spike propagation. By abstracting from the original Hodgkin and Huxley framework and supporting it with contemporary neuromorphic research, this model serves as the conceptual blueprint for the hardware implementation developed in Chapter 4. Each subsection builds upon the previous, ensuring a structured and logical decomposition of neural functionality into circuit-level components.

3.1 The Hodgkin–Huxley Foundation and Cable Theory Abstraction

The Hodgkin and Huxley (HH) model remains a foundational contribution to neuroscience, accurately describing the ionic mechanisms underlying the action potential in the squid giant axon [6]. It models the cell membrane as a nonlinear electrical circuit comprising variable conductance for sodium and potassium, a leakage current, and a membrane capacitance. These time-dependent ion channels regulate excitability and determine the precise timing and shape of spikes.

Figure 3.1.1 provides a hybrid anatomical and electrical representation of a neuron based on the Hodgkin–Huxley model, illustrating how the ionic mechanisms at the membrane interface can be described using voltage-dependent conductance terms.

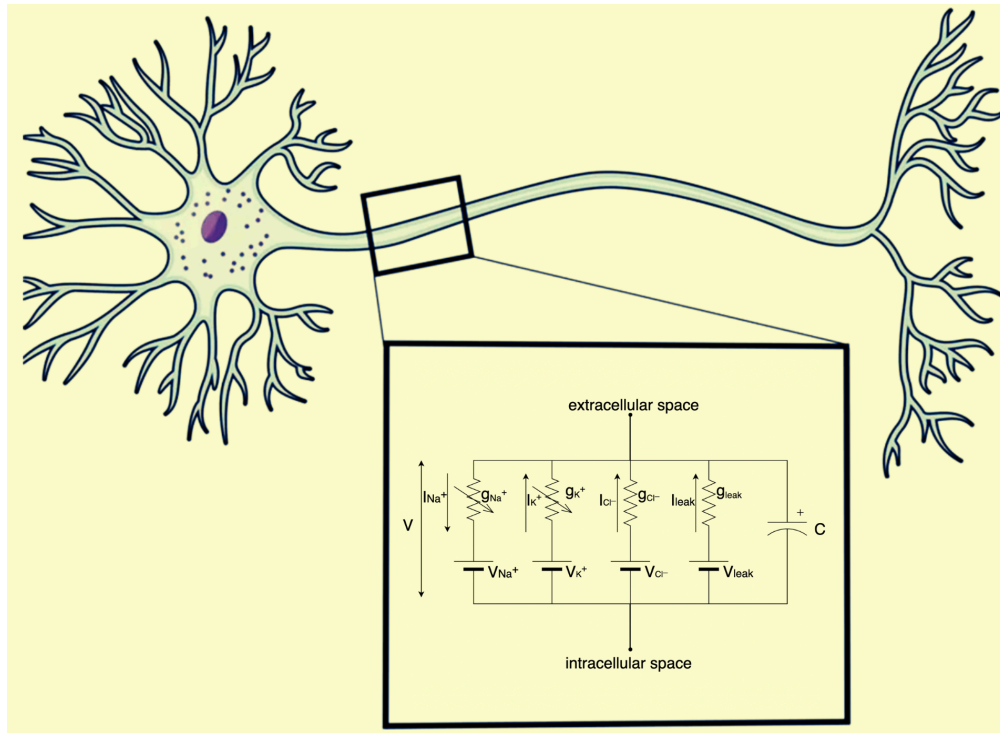


Figure 3.1.1: A hybrid anatomical and electrical representation of a neuron based on the Hodgkin-Huxley model, illustrating ion conductance (Na^+ , K^+ , Cl^- , leak) as circuit components across the membrane [33].

While biologically accurate, the full HH model involves complex differential equations and voltage-gated kinetics that are computationally expensive and often unnecessary for circuit-level abstractions. Therefore, in this project, the HH model is used not for direct replication but to guide the abstraction of the neuron's functional behaviour into modular circuit elements. Specifically, the membrane capacitance is preserved as a critical component of signal integration, while the nonlinear channel dynamics are replaced by thresholding behaviour using comparators.

Cable theory extends the HH model by describing how signals propagate across spatially distributed structures such as dendrites and axons. This continuous-space model treats neuronal branches as cylindrical conductors with distributed resistance and capacitance. In this abstraction, the dendritic and axonal structures are effectively passive transmission lines, filtering and delaying signals based on their geometry and electrical properties. In the context of this design, this biological principle is mapped onto cascaded low-pass RC filters, each section emulating the leaky integrative behaviour of dendritic branches [1], [26].

The abstraction of cable theory is crucial not only for dendritic filtering but also for understanding axonal conduction delays. Axons do not transmit signals instantaneously. Rather, they regenerate action potentials at intervals known as Nodes of Ranvier. These nodes act as amplifiers, restoring the signal's strength as it propagates. The same principle is used in this project's axon model, where cascaded amplifying stages maintain spike integrity across transmission blocks.

Figure 3.1.2 summarises this abstraction using a three-level schematic: the biological neuron, the corresponding cable model, and the compartmental circuit model implemented in hardware.

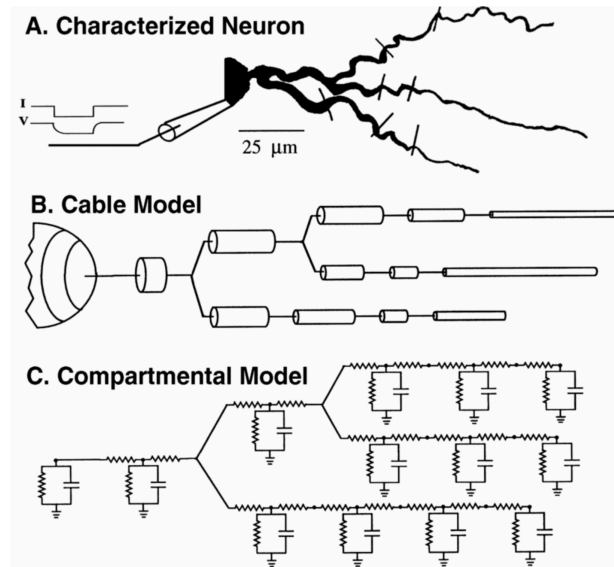


Figure 3.1.2: Illustration of neuron abstraction techniques, showing (A) a characterized biological neuron, (B) the corresponding cable model representing spatial dendritic signal propagation, and (C) its equivalent compartmental model using distributed RC components. Adapted from [34].

This modular abstraction of HH and cable theory ensures that biologically meaningful functionality is preserved while maintaining compatibility with analog circuit constraints. The resultant model is computationally efficient, mathematically stable, and implementable using standard analog components.

3.2 Dendritic Filtering via RC Networks

Dendrites are the primary recipients of synaptic inputs, and their morphology and passive properties significantly influence neuronal output. Biologically, dendrites integrate incoming signals both temporally and spatially, with electrotonic decay reducing signal amplitude as a function of distance from the soma [1]. This filtering behaviour can be approximated using RC filters in circuit design.

The transfer function of a first-order RC filter is:

$$H(S) = \frac{1}{1 + sRC}$$

with a cutoff frequency:

$$f_c = \frac{1}{2\pi RC}$$

Where,

- R is the resistance in ohms (Ω),
- C is the capacitance in farads (F),
- f_c is the cutoff frequency in Hz,
- $\pi \approx 3.1416$

- s is the complex frequency variable,
- $H(S)$ describes how the input signal is modified in frequency domain.

This relationship directly guides component selection. In biological dendrites, filtering is not uniform. Proximal dendrites integrate fast-changing signals, while distal dendrites integrate slower inputs due to larger electrotonic length constants. To mimic this heterogeneity, multiple RC branches with different RCRCRC values can be connected in parallel. The resulting circuit demonstrates temporal selectivity, where each branch captures different aspects of the input waveform [26].

This approach ensures that dendritic filters pass the alpha band (8–13 Hz) while suppressing high-frequency transients and irrelevant oscillations. In practice, MATLAB Bode plots were used to confirm that the design reliably attenuates noise while maintaining the desired temporal fidelity.

3.3 Soma Threshold Mechanism Using Comparator Circuits

The soma is responsible for converting graded dendritic inputs into discrete spikes once the signal exceeds a critical threshold. In the proposed model, this functionality is implemented using a comparator (LM311) followed by a fixed Schmitt trigger (74HC14) and an inverter (74HC04). Together, these components replicate the transition from analog signal summation to all-or-none digital-like spiking, a behaviour fundamental to biological neurons [6], [12].

The reference voltage applied to the comparator is defined by a resistive divider as:

$$V_{ref} = V_{cc} \cdot \frac{R_2}{R_1 + R_2}$$

Where,

- V_{ref} is the comparator threshold voltage in volts (V),
- V_{cc} is the supply voltage in volts (V),
- R_1, R_2 are the divider resistances in ohms (Ω).

The LM311 continuously compares the dendritic output with V_{ref} . When the input voltage surpasses this threshold, the comparator switches high, initiating spike formation. The output is then sharpened by a Schmitt trigger, which introduces hysteresis to suppress small voltage fluctuations and ripple-induced false triggering. This behaviour is analogous to the refractory mechanism in biological neurons, where a minimum depolarisation must be reached before a subsequent spike can occur [6], [12].

Finally, an inverter gate (74HC04) restores the correct output polarity, ensuring that low-to-high dendritic crossings result in well-defined digital spikes. This combination of comparator thresholding and Schmitt-trigger noise immunity provides a robust hardware abstraction of Hodgkin–Huxley threshold dynamics [3], [4]. The result is a faithful circuit-level mechanism for spike initiation that preserves timing fidelity and produces consistent 0–4 V pulses suitable for downstream axonal transmission.

3.4 Axonal Conduction and Signal Regeneration

Once a spike is generated at the soma, it must propagate along the axon with minimal distortion. In biological neurons, this is achieved by regenerative conduction at the Nodes of Ranvier, where voltage-gated ion channels periodically restore signal amplitude [6]. Without such regeneration, spikes would attenuate due to membrane capacitance and axial resistance. The analog model replicates this behaviour using cascaded RC sections combined with periodic amplification and spike shaping circuits [19], [24].

The width of each spike is standardised using a monostable pulse generator, which produces a uniform output pulse regardless of input variability. The pulse duration is given by:

$$T_p = 0.693 R_M C_M$$

Where,

- T_P is the spike pulse width in seconds (s),
- R_M is the monostable timing resistance in ohms (Ω),
- C_M is the monostable timing capacitance in farads (F).

Each axonal section introduces a time constant:

$$\tau_{sec} = R_s C_s$$

Where,

- τ_{sec} is the per-section time constant in seconds (s),
- R_s is the section resistance in ohms (Ω),
- C_s is the section capacitance in farads (F).

The time constant defines both the delay and the extent of attenuation in each segment. To ensure pulse fidelity, τ_{sec} is constrained to remain below one-fifth of the spike width, preventing excessive spreading of the waveform.

The amplifier is implemented as a non-inverting stage, with gain:

$$A_v = 1 + \frac{R_f}{R_{in}}$$

Where,

- A_v is the amplifier gain,
- R_f is the feedback resistance in ohms (Ω),
- R_{in} is the input resistance in ohms (Ω).

This modular regenerative approach ensures that spike amplitude, width, and timing are preserved across the axonal pathway. The use of discrete RC sections with periodic amplification mirrors the function of biological Nodes of Ranvier, where action potentials are re-amplified to maintain fidelity during propagation [24]. The model therefore demonstrates that long-distance spike conduction can be realised in analog hardware using simple, reproducible building blocks.

3.5 Rationale Behind Component Selection

The neuron model was constructed using standard analog components, including RC networks, operational amplifiers, comparators, and Schmitt triggers. These devices were selected because they provide a balance between biological realism, simplicity, and practical feasibility. RC networks capture the leaky integrative behaviour of dendrites and axons, while operational amplifiers restore attenuated signals and maintain stability across cascaded transmission stages. Comparators such as the LM311 replicate the soma's function as a threshold detector, producing discrete spikes when dendritic inputs exceed a critical level. The use of Schmitt triggers, with their built-in hysteresis, ensures noise immunity and prevents false switching, thereby reinforcing the reproducibility of spike generation.

By relying exclusively on widely available and well-characterised components, the design avoids the fabrication challenges and variability associated with emerging technologies such as memristors or floating-gate transistors [17], [20], [25]. At the same time, the architecture remains compatible with future upgrades, leaving scope for the introduction of adaptive or plastic behaviours. The use of these standard components ensures that the model is cost-effective, reproducible, and aligned with both biological principles and engineering feasibility.

3.6 Simulation Tools: Multisim and MATLAB

The simulation and validation of the proposed analog neuron model were carried out using two primary tools: Multisim and MATLAB. These platforms were selected based on their ability to evaluate different aspects of circuit performance. Multisim provided an environment for time-domain circuit behaviour analysis, while MATLAB offered frequency-domain insight through transfer function and signal response validation. This dual-tool methodology ensured both biological plausibility and circuit reliability.

3.6.1 Multisim for Time-Domain Circuit Behaviour

Multisim served as the primary circuit simulation platform for evaluating the dynamic behaviour of individual neuron subsystems. It was used to construct and test the RC-based dendritic filters, the comparator-based soma threshold circuit, the Schmitt trigger pulse shaper, and the monostable pulse generators and amplifier blocks used in axonal modelling.

The tool enabled visual and quantitative monitoring of voltage and current across various nodes, allowing direct observation of how input signals evolved as they passed through the system. Multisim enabled assessment of comparator threshold crossings, pulse generation timing, and the integrity of signal propagation across a nine-stage axonal block. The Schmitt trigger stages were validated for consistency in pulse width, amplitude, and delay characteristics. The simulation environment closely matched the physical hardware components used, such as LM311 comparators and 74HC14 Schmitt triggers. This ensured that circuit behaviour in simulation was predictive of physical implementation outcomes.

Multisim was selected for its component library, ease of prototyping with off-the-shelf parts, and compatibility with educational and laboratory workflows. Although more advanced tools such as LTspice or Cadence Spectre offer finer control over IC-level simulations, they were not required for this discrete analog implementation. Instead, Multisim provided an optimal balance between schematic-level modelling, real-time observation, and hardware-aligned component behaviour.

3.6.2 MATLAB for Frequency-Domain Validation

To complement time-domain analysis, MATLAB was used for high-precision frequency-domain validation of the dendritic RC filters and spike propagation behaviour. Using analytical transfer function modelling, MATLAB enabled quantitative assessment of the circuit's filtering properties.

Simulations demonstrated that the filters reliably attenuated high-frequency noise while maintaining temporal fidelity for lower-frequency biological signals. These findings supported the biological realism of the dendritic stage and confirmed that irrelevant signal content was suppressed before reaching the soma threshold circuit.

MATLAB was also used to simulate axon pulse propagation through the nine-stage monostable-amplifier blocks. The focus was on visualising signal integrity, pulse width preservation, and delay across stages. By applying Fast Fourier Transform (FFT) to the generated pulses, it was confirmed that the system did not distort the frequency content of spikes during propagation. This mirrored the regenerative function of Nodes of Ranvier in biological axons.

3.6.3 Complementarity of Tools

The decision to use both Multisim and MATLAB was driven by their complementary strengths. Multisim allowed direct interaction with the schematic design and provided realistic transient simulations aligned with physical hardware behaviour. However, its frequency-domain capabilities were limited. MATLAB addressed this gap by offering a mathematically precise environment to analyse filter performance, transfer function dynamics, and FFT-based signal verification.

Together, these tools validated the model across both time and frequency domains. The use of Multisim ensured correct signal generation and shape at each stage, while MATLAB provided assurance that signal filtering and propagation conformed to theoretical expectations and biological constraints. This dual-validation strategy strengthened confidence in the model's readiness for physical implementation.

3.7 Assumptions, Idealities, and Constraints

As with any hardware modelling of a biological system, the analog neuron design required several simplifying assumptions and idealizations. These decisions were necessary to maintain feasibility while preserving the core electrophysiological characteristics of real neurons.

3.7.1 Assumptions:

- It is assumed that all dendritic signals are analog voltage waveforms, and their summation can be represented using Kirchhoff's current law at a node.
- The axon is modelled as a sequence of identical transmission and regeneration blocks. Biological variation in signal delay or amplitude due to ion channel dynamics is not explicitly modelled.

- Biological action potentials are assumed to be effectively captured by standardized monostable pulses of fixed width and amplitude. This simplification allows for consistent propagation but ignores ionic recovery dynamics.
- Comparator thresholds are treated as static values, whereas real neurons exhibit dynamic threshold adaptation over time.

3.7.2 Idealities:

- Passive components are assumed to behave ideally, with no parasitic inductance, temperature drift, or tolerance variation. In practice, resistor and capacitor values can deviate by 1 to 5 percent.
- Op-amps are treated as having infinite gain and bandwidth, instantaneous slew rate, and zero offset voltage. While real op-amps exhibit limitations, ideal behaviour simplifies modelling and ensures predictable performance.
- Power supply noise and electromagnetic interference are neglected in simulation. These may influence real-world output fidelity.

3.7.3 Constraints:

- The model is constructed using only idealised circuit elements within a simulation environment, which limits the ability to capture real-world electrical noise, thermal fluctuations, and hardware-specific non-linearities.
- The dendritic and axonal pathways are represented using simplified linear cascades, rather than biologically accurate branched morphologies. This reduction enables controlled analysis but sacrifices spatial complexity.
- All parameter values, including comparator thresholds, time constants, and pulse widths, are fixed during simulation. This prevents the model from reflecting plasticity or adaptive behaviours observed in real neurons.
- The simulation environment assumes perfect component behaviour and interconnections. Effects such as impedance mismatches, parasitic couplings, or environmental interference are excluded, which may limit applicability to physical circuit implementations.

3.7.4 Comparison with Biological Reality

The following table summarises the assumptions and idealised behaviours for key circuit components, mapped to their biological analogues. The final column highlights trade-offs introduced by each abstraction.

Component / Element	Biological Counterpart	Assumption in Model	Trade-off / Implication
Resistor (R)	Ion channel leak conductance	Assumed to be linear, temperature-independent, and precise	Real ion channels have non-linear, voltage-dependent conductance; model does not capture adaptive dynamics.
Capacitor (C)	Cell membrane capacitance	Treated as ideal with no parasitics or tolerance variation	Biological membranes show nonlinear capacitance; real capacitors may drift with temperature or age.
Op-Amp	Sodium-Potassium ion channel dynamics	Considered to have infinite gain, zero offset, unlimited slew rate and bandwidth	Real op-amps introduce delay and bandwidth limitations; biological channels have complex kinetics.
Comparator Threshold	Neuronal firing threshold (axon hillock)	Treated as static value	Real neurons exhibit dynamic thresholds (e.g., adaptation, fatigue) not modelled here.
Schmitt Trigger Hysteresis	Refractory period modulation	Fixed upper and lower threshold voltages	Biological hysteresis may vary with recent activity or ion concentration.
Cascaded Amplifier Blocks	Nodes of Ranvier for spike regeneration	Assumed ideal amplification without noise or power loss	Real axonal transmission can fail or be affected by metabolic state; signal integrity may degrade.
Signal Summation Node	Soma's synaptic integration point	Assumes perfect linear summation (Kirchhoff's law)	In real neurons, synaptic inputs are modulated by spatial and temporal effects; integration is nonlinear.
Power Supply and Ground	Cellular ionic balance and extracellular environment	Treated as stable and noise-free	Biological systems show variability in extracellular potential and ion supply stability.

Table 3.7: Assumptions and Idealities vs. Biological Realism

This comprehensive mapping illustrates the balance between simplicity and fidelity. While several abstractions are used, they allow the analog model to capture the core behaviours of synaptic integration, spike thresholding, and axonal propagation. The design decisions discussed here ensure that the model remains biologically inspired, analytically traceable, and practically implementable using widely accessible tools and components.

3.8 Summary of Theoretical Architecture

The complete theoretical neuron model developed in this chapter serves as a biologically motivated and hardware-feasible framework for analog implementation. The system is modular, consisting of clearly defined blocks that mimic the key stages of neural signal flow: input reception and integration (dendrites), threshold-based decision-making (soma), and signal regeneration and transmission (axon).

Dendritic behaviour is emulated through RC low-pass filters that selectively allow biologically relevant frequencies to pass. This stage ensures that only valid inputs reach the decision node. The soma is represented by a comparator (LM311), which functions as a static threshold detector. To replicate all-or-none spiking behaviour, this comparator output feeds into a Schmitt trigger (74HC14), which introduces hysteresis and generates sharp digital spikes.

The axon pathway is realised using a 9-block RC transmission line, where each block introduces temporal delay and attenuation, modelling axonal conduction characteristics. Signal strength is periodically restored using amplifiers placed after every three blocks. These amplifier stages mimic the function of Nodes of Ranvier, enabling signal regeneration over extended lengths without degradation. The final output represents the spike signal at the axon terminal.

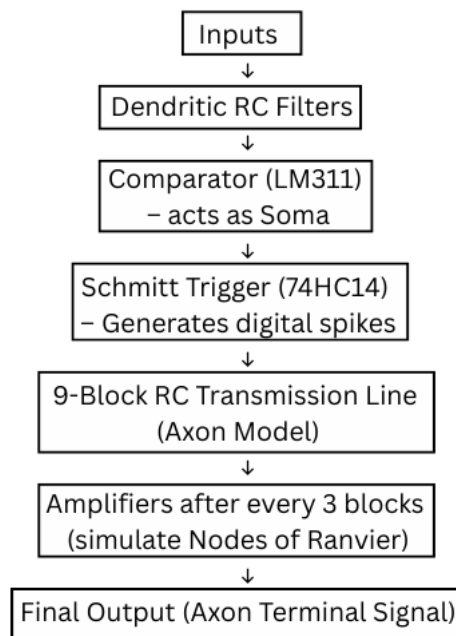


Figure 3.8: Functional Block Diagram of the Analog Neuron Model

This diagram encapsulates the full signal flow from input to output, passing through dendritic filters, threshold circuits, digital spike generation, and axonal conduction with periodic amplification. The use of modular blocks allows for clarity, repeatability, and targeted validation of each processing stage.

The component choices, including op-amps and discrete RC elements, ensure that the model remains cost-effective and reproducible using standard academic tools. MATLAB is used to verify the frequency response of RC filters, while Multisim ensures time-domain validation of circuit behaviour.

Assumptions such as ideal op-amp response, static comparator thresholds, and ideal power conditions are acknowledged. Limitations due to fixed RC values and simplified topology are also considered. Nevertheless, the model successfully captures essential biological behaviours while remaining tractable for analog hardware design.

This model forms the foundation for the physical implementation explored in the next chapter.

Chapter 4: Circuit Implementation and Simulation Results

This chapter presents the complete implementation of the proposed analog neuron model at the circuit level. Each section focuses on a distinct stage of the neuron, including input signal design, dendritic signal filtering, soma-based spike generation, and axonal transmission with amplification. All simulations were carried out using a combination of MATLAB and Multisim to validate the model's functional accuracy and biological relevance.

To provide a clear overview of the overall system architecture, Figure 4.1 introduces a consolidated visual mapping between the biological neuron and its corresponding electronic implementation. The diagram illustrates how each anatomical region, such as dendrites, soma, axon, and the node of Ranvier, has been represented using circuit blocks. It also highlights the flow of signal from external neural inputs through individual processing stages, culminating in spike generation and axonal propagation. This figure serves as a useful reference point for understanding the placement and function of each subcircuit discussed in the subsequent sections.

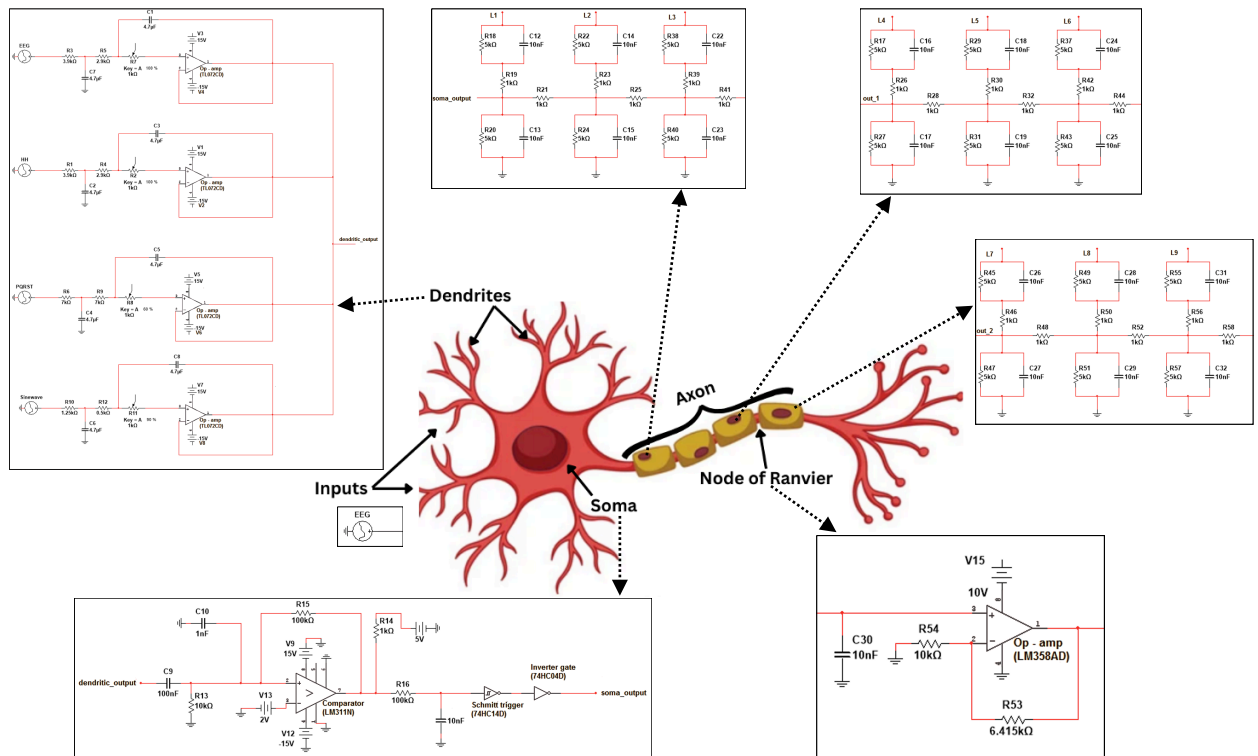


Figure 4.1: Mapping of Biological Neuron to Analog Circuit Implementation

4.1 Input Signal Design

The analog neuron circuit was tested using four biologically inspired input waveforms, each designed in MATLAB. These signals were selected not only for their relevance to real neural or physiological activity but also to introduce deliberate complexity. The purpose was to simulate realistic input scenarios and rigorously evaluate the ability of the dendritic filter stage to preserve relevant information while suppressing irrelevant or high-frequency components. Each waveform was scaled to lie within ± 50 mV or ± 70 mV to ensure compatibility with analog components in the circuit.

4.1.1 Voltage Unit Convention

All input signals were designed in MATLAB using biologically plausible amplitude ranges, typically within ± 50 millivolts or ± 70 millivolts. These values reflect the membrane potential fluctuations observed in real neurons and were selected to preserve the biological grounding of the model during input construction.

Before applying the signals to the circuit simulation, the waveforms were scaled to volt-level ranges. This adjustment was introduced intentionally during the signal preparation phase, prior to the axon stage (section – 4.3). The purpose of this scaling

was to ensure compatibility with analog components in Multisim, particularly to allow accurate thresholding and spike generation within voltage domains that the simulation environment handles efficiently. This conversion avoids numerical limitations, ensures reliable comparator operation, and preserves the intended signal dynamic [18], [19], [23].

This approach is valid for analog neuron modelling, as the circuit's behaviour depends on the shape and timing of the signals rather than on their absolute voltage levels. The resistor-capacitor parameters in the dendritic filters, and the comparator thresholds in the soma and axon stages, were tuned accordingly to operate within this volt-level domain.

Throughout this chapter, analog and subthreshold waveforms such as the EEG, HH, and PQRST signals are labelled in millivolts to reflect their original biological interpretation. Comparator-based outputs and spike waveforms from the axon stage are labelled in volts to indicate their digital switching nature.

4.1.2 EEG Input

The EEG waveform was extracted from the PhysioNet open-access database [31]. To introduce complexity and test the dendritic filter's selectivity, additional components were added in MATLAB. These included dense sine waves spanning 5–90 Hz to simulate background neural interference and sparse high-frequency transients from 100–450 Hz to mimic artifacts. Gaussian noise was also added to simulate synaptic variability and real-world recording noise. The final waveform was amplitude-normalized to ± 70 mV. This signal was used to assess the circuit's ability to preserve low-frequency neural oscillations while attenuating overlapping noise and high-frequency distortion.

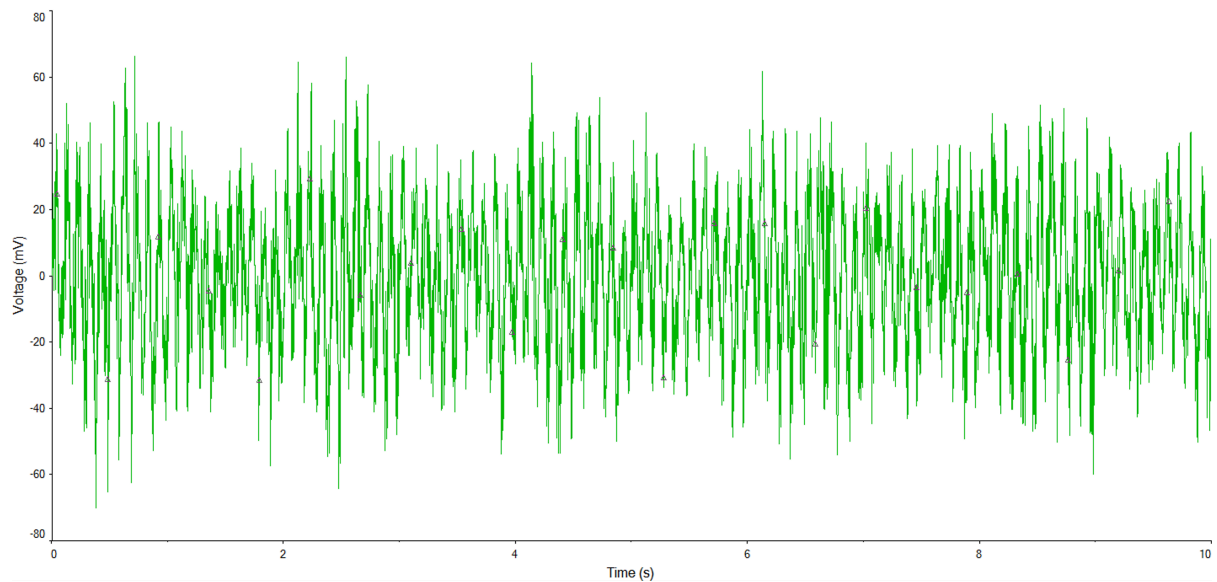


Figure 4.1.1: Time-domain plot of the EEG input signal.

Figure 4.1.1 displays the time-domain plot of the simulated EEG signal. The waveform exhibits low-frequency oscillatory dynamics embedded in a background of high-frequency disturbances and Gaussian noise. The voltage range remains confined to ± 70 mV, reflecting realistic extracellular recording levels. The signal's amplitude variability and non-stationarity mimic typical neural signals encountered in cognitive or resting-state experiments.

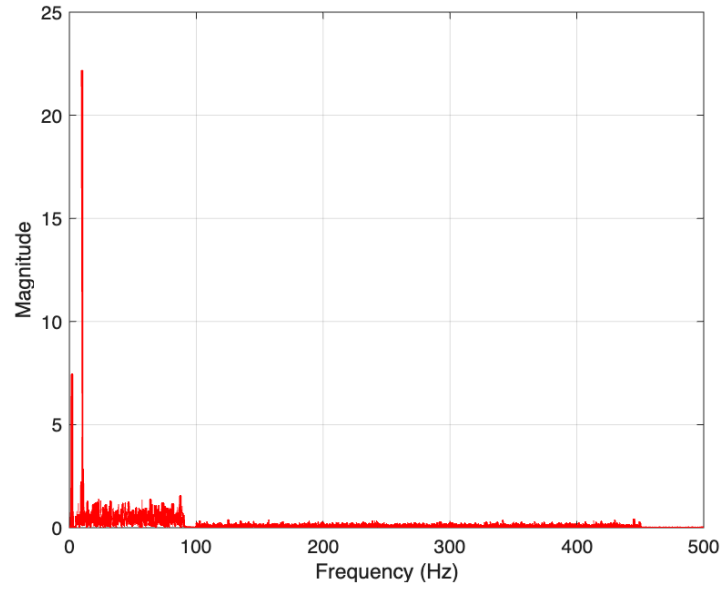


Figure 4.1.2: Frequency-domain representation of the simulated EEG signal.

Figure 4.1.2 shows the corresponding frequency-domain representation using the Fast Fourier Transform (FFT). The spectrum reveals a concentration of energy below 50 Hz, which is characteristic of natural EEG rhythms. Additional high-frequency components reflect the intentionally introduced noise and artifacts, used to evaluate the dendritic circuit’s ability to preserve meaningful oscillatory activity while suppressing irrelevant frequencies.

4.1.3 Hodgkin–Huxley Spike Train

This input was inspired by the classical Hodgkin–Huxley model, which defines the ionic basis of action potentials using differential equations. A train of Gaussian-shaped depolarising spikes was generated at intervals to mimic typical neuronal firing patterns. To introduce complexity, the waveform was modulated with a slow envelope, overlaid with dense sine-based interference and Gaussian noise. The amplitude was scaled to ± 50 mV to match intracellular voltage levels and circuit compatibility. This input tested the system’s ability to preserve fast, transient spikes within a noisy background.

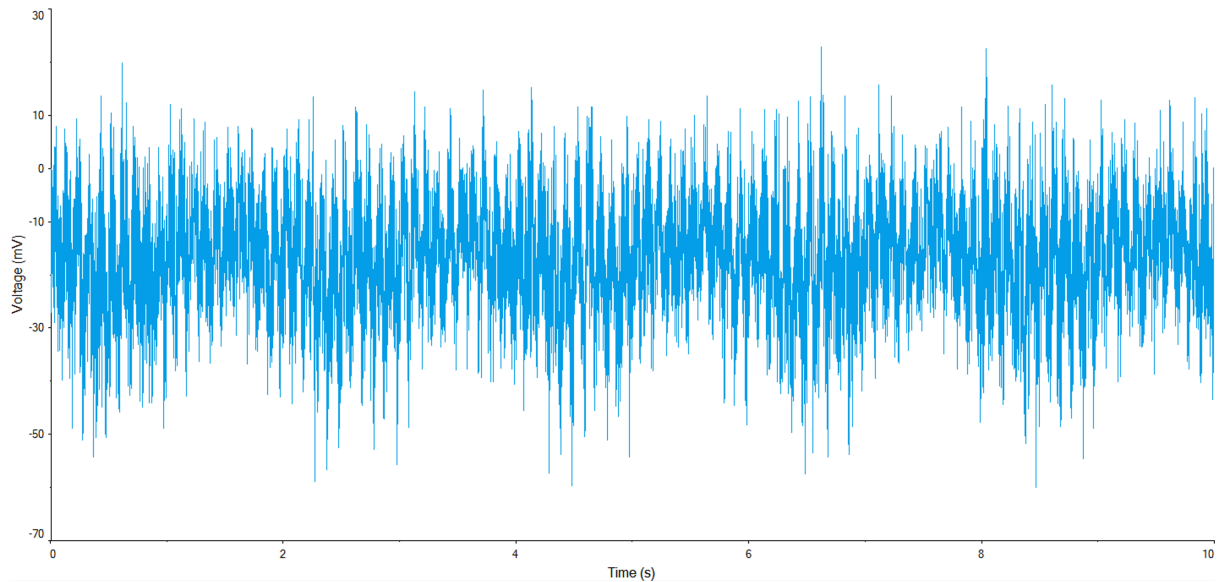


Figure 4.1.3: Time-domain representation of the HH input signal.

Figure 4.1.3 illustrates the time-domain waveform of the HH input. The waveform captures sharp, transient spikes riding on a fluctuating baseline. The spikes are irregular in amplitude and spacing due to the presence of noise and modulation, thereby mimicking realistic firing activity with subthreshold fluctuations.

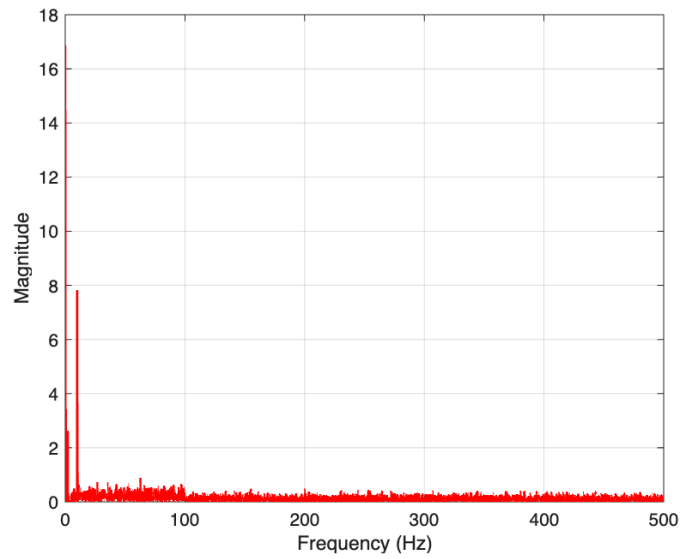


Figure 4.1.4: Frequency-domain representation of HH signal.

Figure 4.1.4 presents the frequency-domain spectrum of the HH signal. The dominant energy lies in the low- to mid-frequency range (<30 Hz), with a spectral tail that extends to higher frequencies due to the spiking transients. The broader spread validates the complexity of the input and the need for targeted filtering to retain functionally relevant dynamics.

4.1.4 PQRST Input

The PQRST input was constructed by synthesising a sequence of Gaussian-shaped pulses representing the P wave, QRS complex, and T wave phases of the cycle. The waveform was repeated periodically across the 10-second simulation window. Additional complexity was introduced using dense background fluctuations, high-frequency transients, and Gaussian noise. The waveform was sampled at 10 kHz and scaled to ± 50 mV. This input served as a test case for structured multi-phase inputs with overlapping interference.

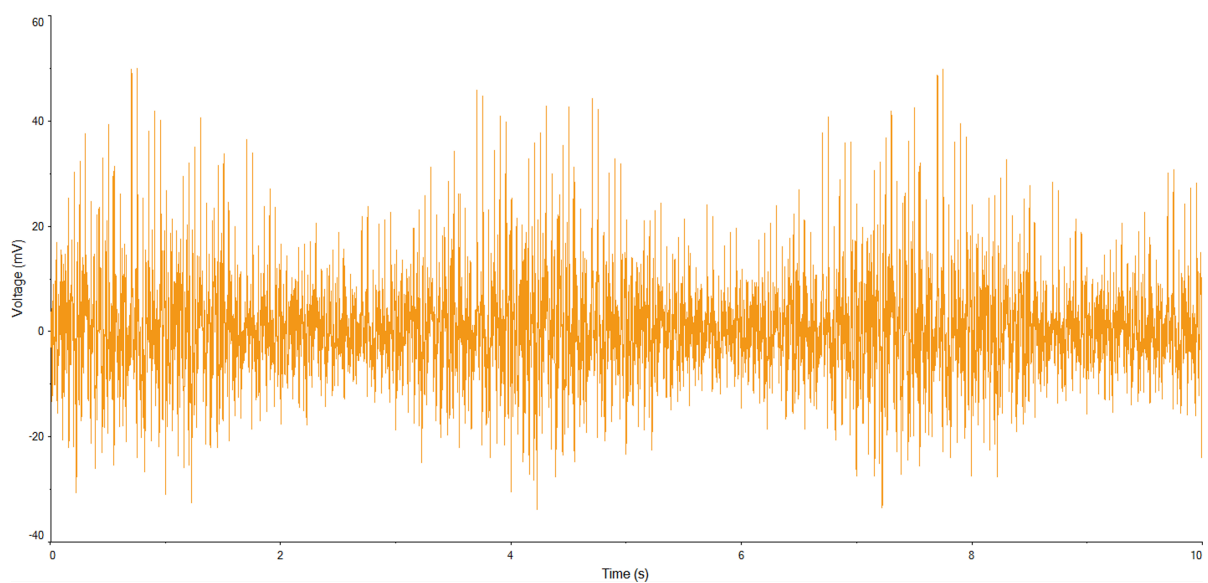


Figure 4.1.5: Time-domain waveform of the PQRST input signal.

Figure 4.1.5 shows the time-domain plot of the PQRS signal. The waveform exhibits structured, multi-phase oscillations typical of cardiac electrical activity, with sharp depolarising peaks and rounded repolarisation phases. Noise components and fluctuations are evident throughout the signal, simulating real-world signals contaminated by movement or external interference.

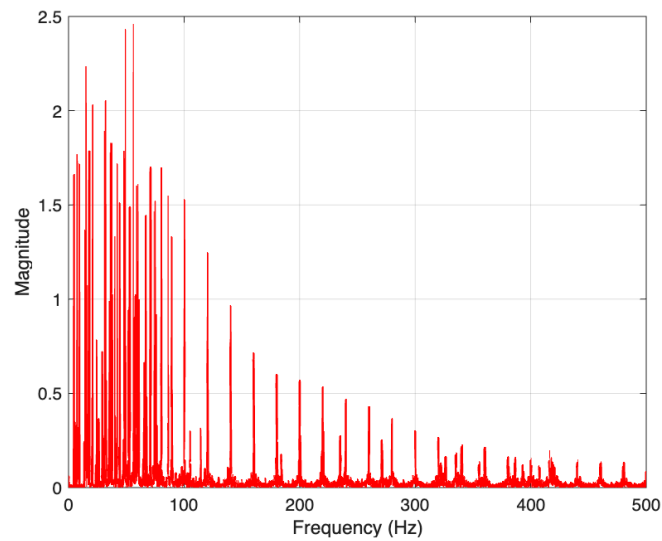


Figure 4.1.6: Frequency-domain representation (FFT) of the PQRS signal.

Figure 4.1.6 presents the FFT of the PQRS signal. The frequency-domain plot reveals a wide distribution of energy across low and mid-range bands, with sharp features translating into broader spectral coverage. The dense spectrum is representative of signals where multiple physiological phases overlap, making this input ideal for testing frequency-selective filter behaviour.

4.1.5 Multi-Sinusoidal Input

The final test signal consisted of a clean summation of three pure sinusoids at 5 Hz, 20 Hz, and 50 Hz. No noise, modulation, or distortion was added. This waveform was used as a control input to verify basic filtering behaviour of the dendritic stage. The signal was scaled to ± 50 mV and sampled at 2 kHz. Because of its simplicity and well-defined frequency content, this input provided a useful reference for understanding the behaviour of the circuit under ideal conditions.

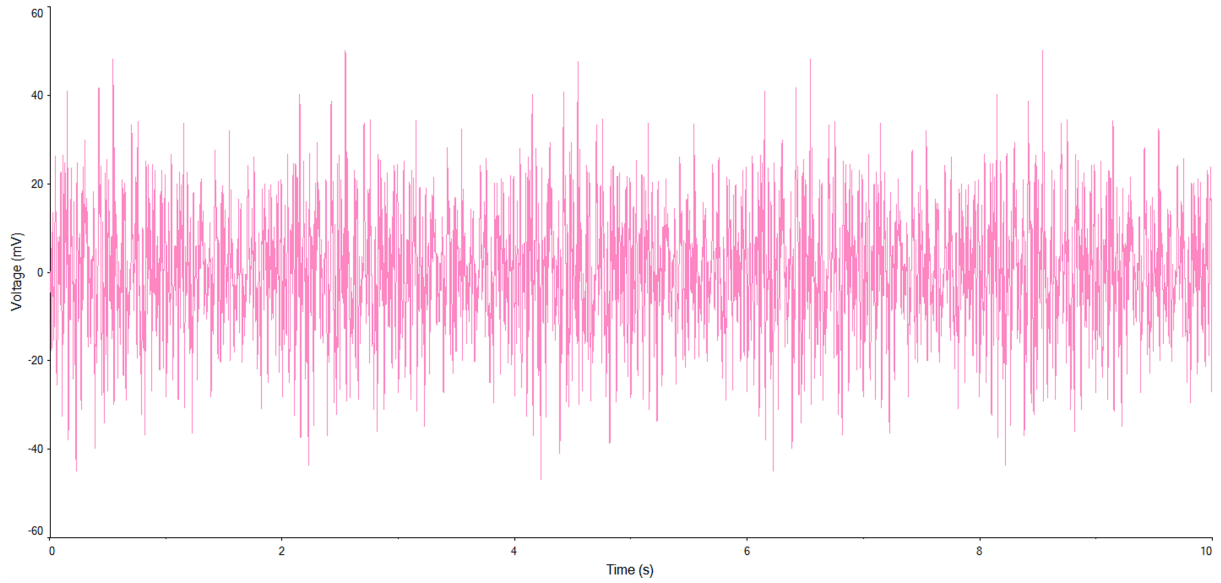


Figure 4.1.7: Time-domain representation of the multi-sinusoidal input signal.

Figure 4.1.7 displays the time-domain waveform of the multi-sinusoidal signal. The waveform demonstrates continuous oscillatory behaviour with visibly superimposed frequencies of differing periodicities. The lack of noise and distortion makes the signal ideal for evaluating the response of the circuit in the absence of real-world complexities.

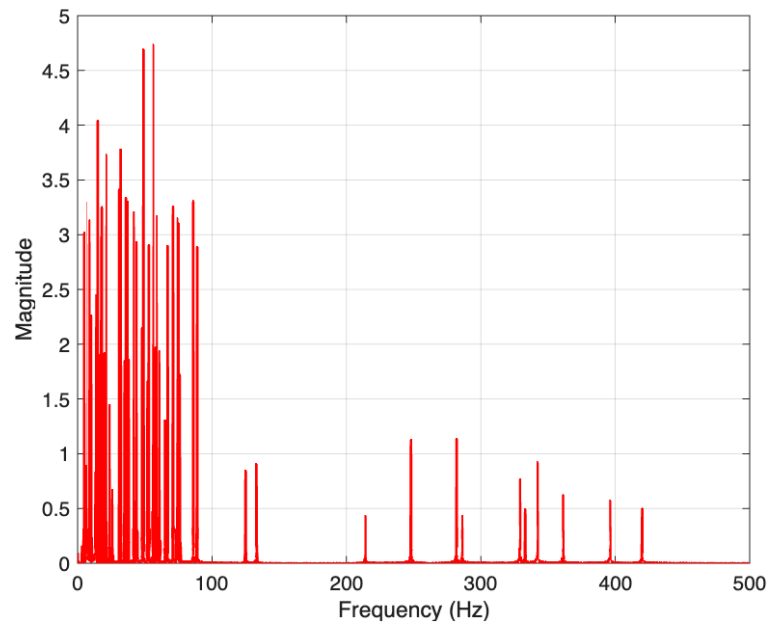


Figure 4.1.8: Frequency spectrum of the multi-sinusoidal input.

Figure 4.1.8 shows the frequency-domain spectrum. The FFT reveals sharp, isolated peaks at 5 Hz, 20 Hz, and 50 Hz, corresponding exactly to the components included in the input. These peaks serve as spectral benchmarks to validate whether the filter circuits are capable of selectively passing desired frequency bands while suppressing others.

4.2 Dendritic Filtering using Tuned Bandpass Circuits

To emulate the frequency-selective behaviour of dendrites, a dedicated bandpass filter was implemented for each of the four distinct input signals: EEG, Hodgkin-Huxley (HH), PQRST, and sinusoidal waveforms. These filters were designed using TL072CD operational amplifiers in a configuration that enables both gain control and frequency shaping. Each filter was

specifically tuned to the alpha frequency band (8–13 Hz), as this range is biologically significant and associated with key neuronal oscillations.

Alpha-band activity is commonly linked to relaxed wakefulness, cortical inhibition, attention regulation, and sensory gating mechanisms. By isolating this band across all input signals, the filters selectively preserve cognitively relevant components while suppressing unwanted noise or irrelevant frequencies. This tuning emulates how dendrites selectively respond to subthreshold inputs that carry functional meaning.

The output from each of these filters was later merged to generate the cumulative dendritic signal, which closely resembles the summation mechanism in biological neurons.

The objective of this stage was twofold:

- (i) to individually condition each input signal by suppressing irrelevant frequency components, and
- (ii) to ensure that only the most meaningful subthreshold signals are passed into the dendritic output for further processing.

4.2.1 Bandpass Filtering of EEG Input

The EEG signal, initially rich in low-frequency components spanning 1–30 Hz, was filtered using an analog bandpass circuit constructed in Multisim. The design (Figure 4.2.1) uses a TL072CD op-amp configured as a non-inverting amplifier. The circuit includes a high-pass RC section at the input, with the lower cutoff frequency defined by resistor and capacitor. Feedback capacitors and resistors shape the upper cutoff boundary. A potentiometer (R7) controls the gain without shifting the passband.

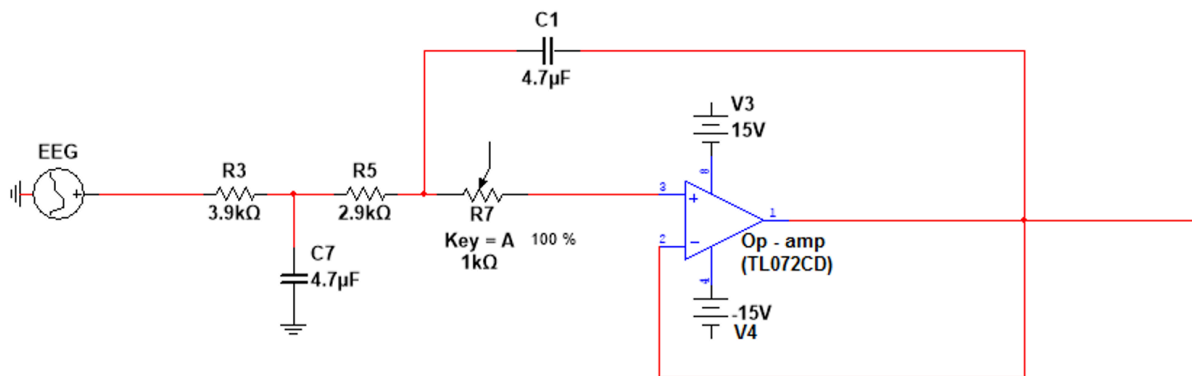


Figure 4.2.1: EEG Bandpass Filter Circuit Design Implemented in Multisim.

Figure 4.2.1 shows the complete EEG bandpass circuit layout. All component values were selected to yield a centre frequency near 10 Hz, aligning with alpha rhythms in the EEG spectrum. The TL072CD's high slew rate and low input bias current ensure faithful analog performance for these biological signals.

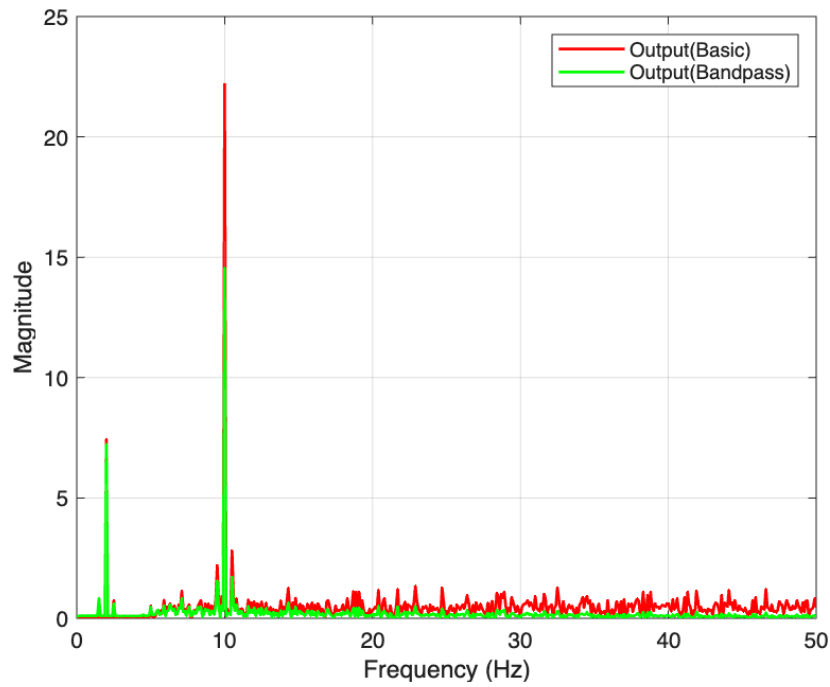


Figure 4.2.2: FFT Comparison of EEG Output Before and After Bandpass Filtering.

The effect of this filter is shown in Figure 4.2.2, where the frequency spectra of the raw EEG output and the filtered output are overlaid. The bandpass response significantly attenuates spectral energy outside the desired 8–13 Hz window, while maintaining the key alpha-band feature around 10.1 Hz.

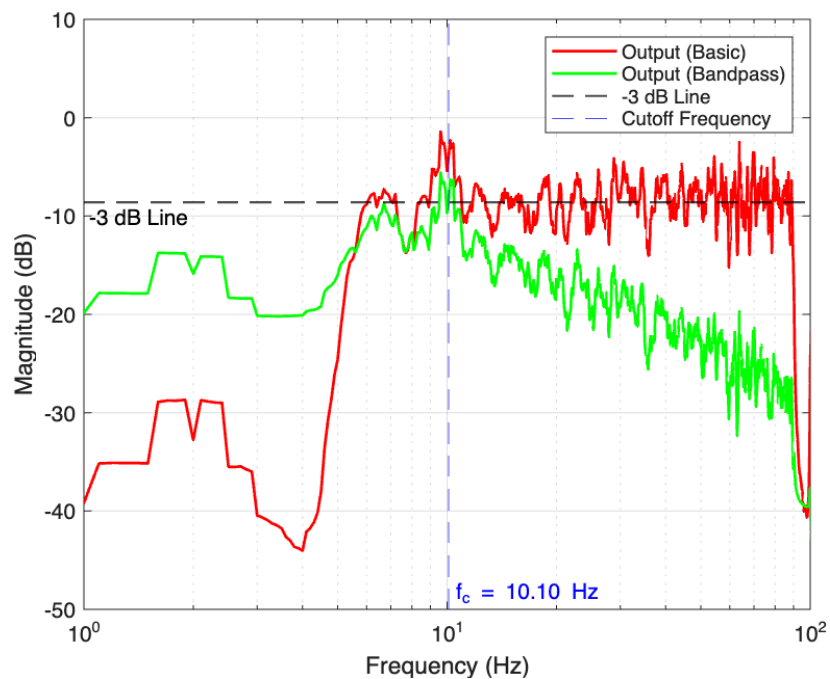


Figure 4.2.3: Bode Magnitude Plot Showing the Cutoff Behaviour of the EEG Filter Circuit.

To validate frequency selectivity, Figure 4.2.3 presents the Bode magnitude plot. A clear -3 dB cutoff is seen near 10.10 Hz, confirming the effectiveness of the filter design in suppressing extraneous low and high frequency components.

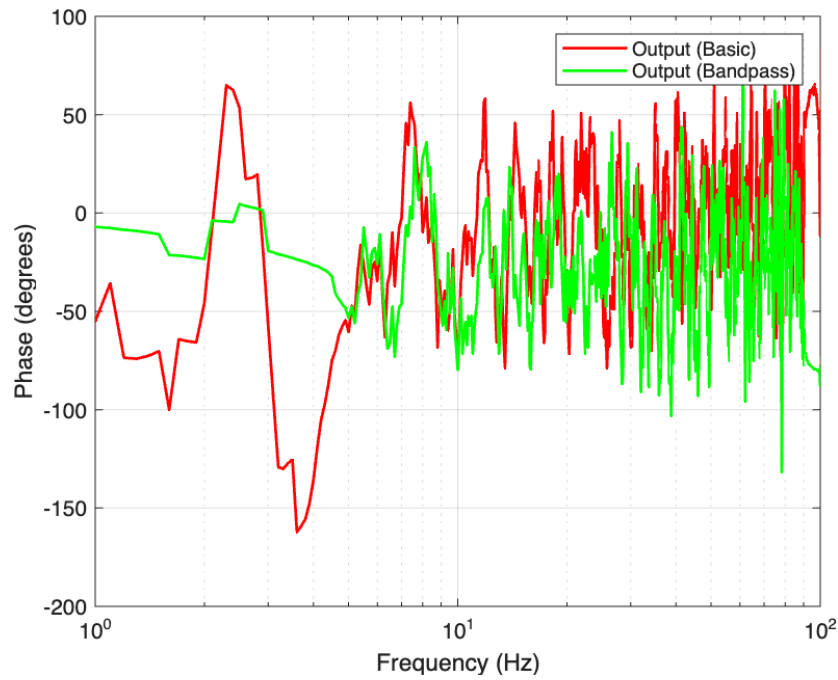


Figure 4.2.4: Bode Phase Plot for EEG Output – Basic vs Bandpass Filtered.

Figure 4.2.4 provides the Bode phase response of the same circuit. A relatively linear phase transition is observed near the passband, indicating minimal distortion and stable timing characteristics for signal propagation into the soma.

4.2.2 Bandpass Filtering of HH Input

The HH waveform represents a more dynamic input characterized by fast transients and wide frequency spread, including both subthreshold and suprathreshold activity. To isolate the subthreshold behaviour, the same bandpass topology was reused (Figure 4.2.5), with identical R and C values for design consistency. Gain was adapted through the potentiometer to ensure amplitude compatibility.

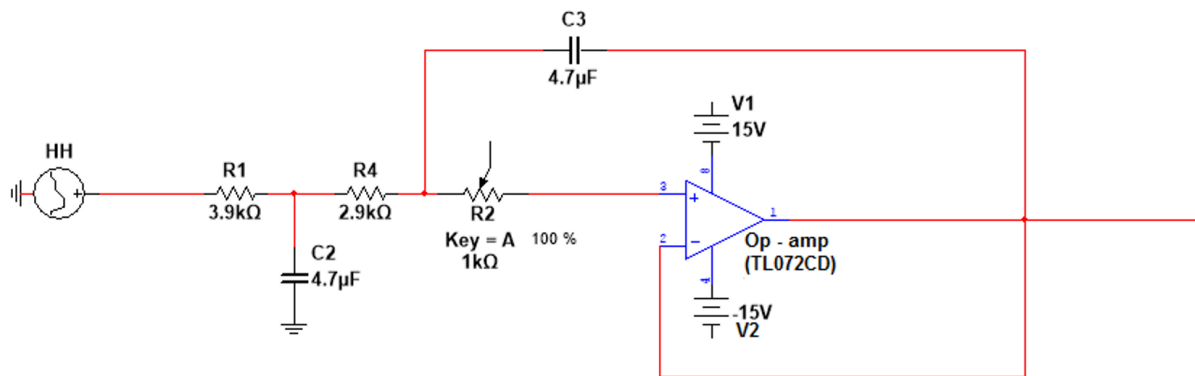


Figure 4.2.5: HH Bandpass Filter Circuit Design Implemented in Multisim

Figure 4.2.5 shows the implemented circuit in Multisim. The high-pass input stage blocks DC content, while the capacitive feedback loop imposes an upper frequency limit. This ensures that the filter suppresses the rapid spiking artifacts while preserving biologically relevant slower dynamics.

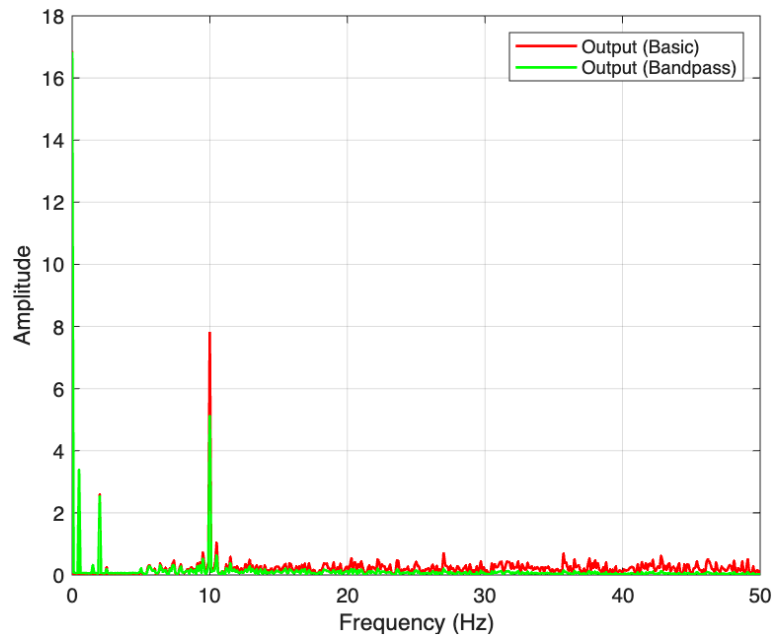


Figure 4.2.6: FFT Comparison of HH Output Before and After Bandpass Filtering.

In Figure 4.2.6, the frequency spectra show that the unfiltered HH signal contains significant high-frequency content which is eliminated after filtering. The remaining dominant spectral peak resides at approximately 10 Hz.

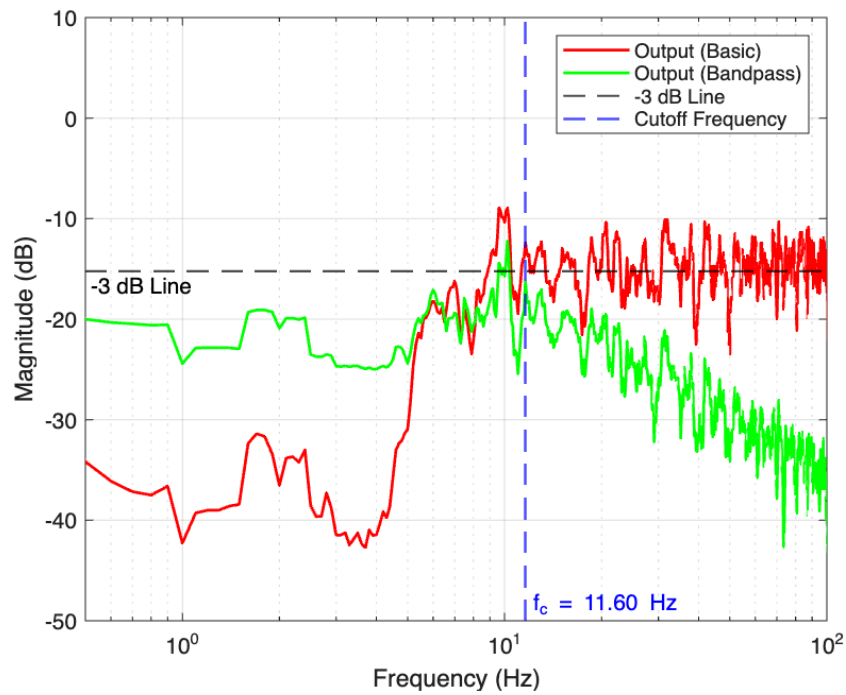


Figure 4.2.7: Bode Magnitude Plot Showing the Cutoff Behaviour of the HH Filter Circuit.

The Bode magnitude plot in Figure 4.2.7 displays a cutoff frequency of 11.60 Hz. This small deviation from the EEG stage is attributed to amplitude differences in the HH signal, potentiometer gain setting, and waveform harmonics. Despite this, the filter still targets the alpha band effectively.

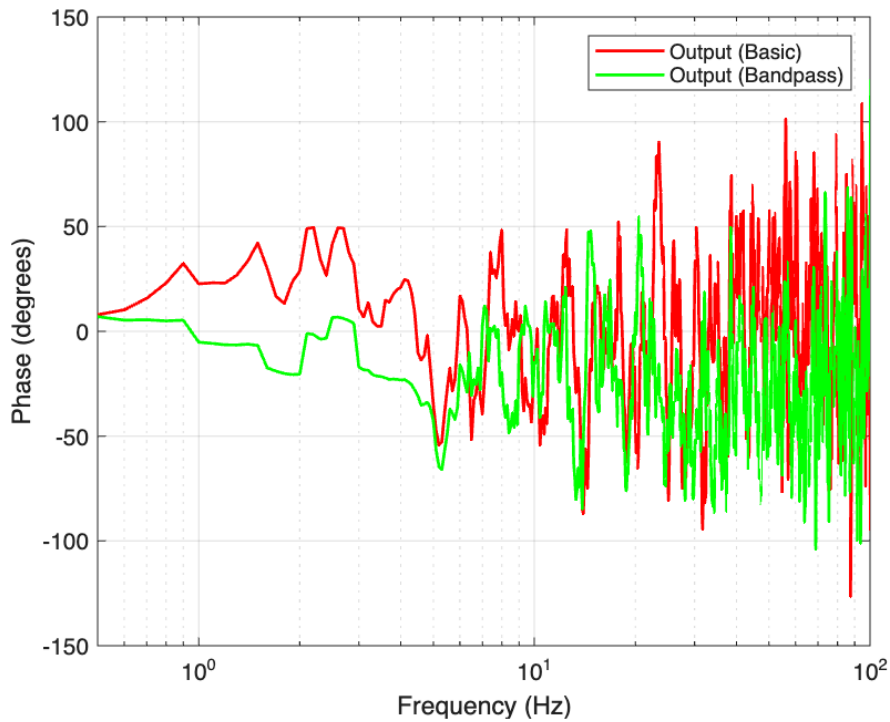


Figure 4.2.8: Bode Phase Plot for HH Output – Basic vs Bandpass Filtered.

Figure 4.2.8 shows the corresponding Bode phase response. The phase curve of the filtered output exhibits a more stable slope within the passband, ideal for synchronized signal summation at the dendritic node.

4.2.3 Bandpass Filtering of PQRST Input

The PQRST signal, generated using pulse-shaped representations of cycles, exhibits multiple overlapping frequency components. The same filtering structure was applied (Figure 4.2.9), with the TL072CD op-amp configured to emphasize the mid-frequency features of the signal.

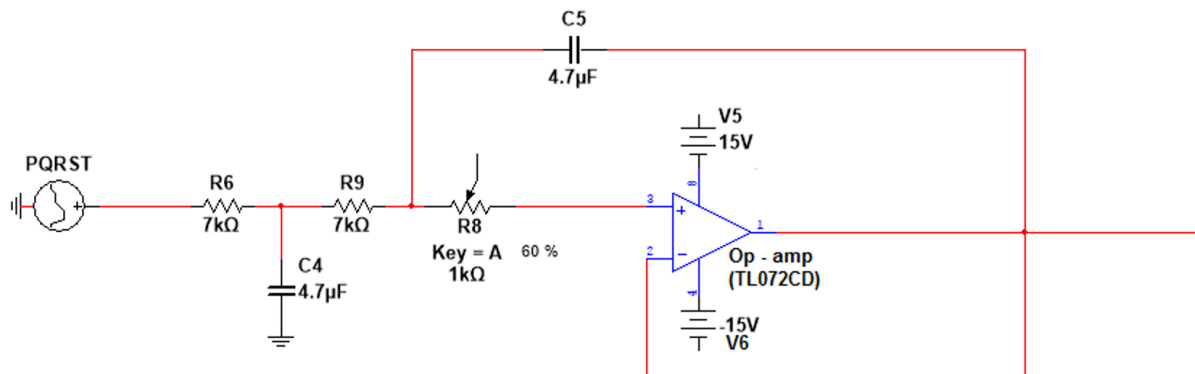


Figure 4.2.9: PQRST Bandpass Filter Circuit Design Implemented in Multisim

Figure 4.2.9 shows the PQRST-specific circuit schematic. The feedback network was slightly tuned to accommodate the broader nature of the PQRST waveform without significantly affecting the target frequency.

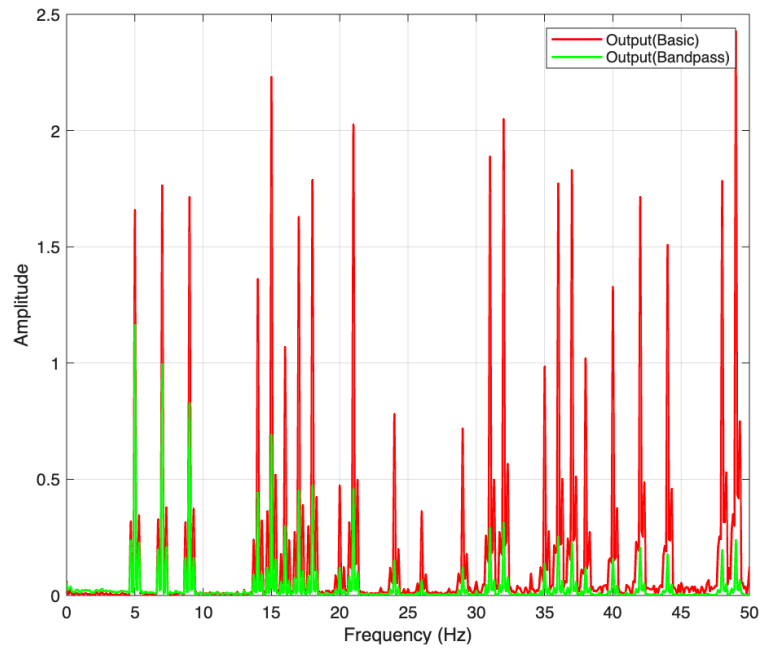


Figure 4.2.10: FFT Comparison of PQRST Output Before and After Bandpass Filtering.

The resulting FFT in Figure 4.2.10 demonstrates that the filter successfully isolates low- to mid-frequency cardiac features around 9 Hz while suppressing DC drift and sharp transients. This enables smooth downstream integration.

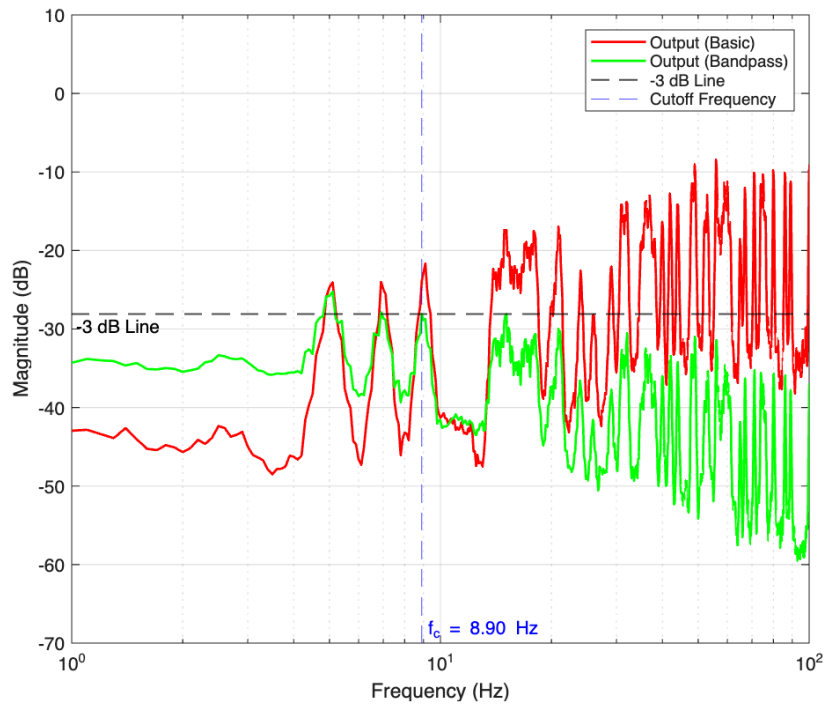


Figure 4.2.11: Bode Magnitude Plot Showing the Cutoff Behaviour of the PQRST Filter Circuit.

The Bode plot in Figure 4.2.11 confirms a -3 dB point around 8.90 Hz. This tuning enhances the biologically meaningful portions of the signal that fall within the alpha-like range.

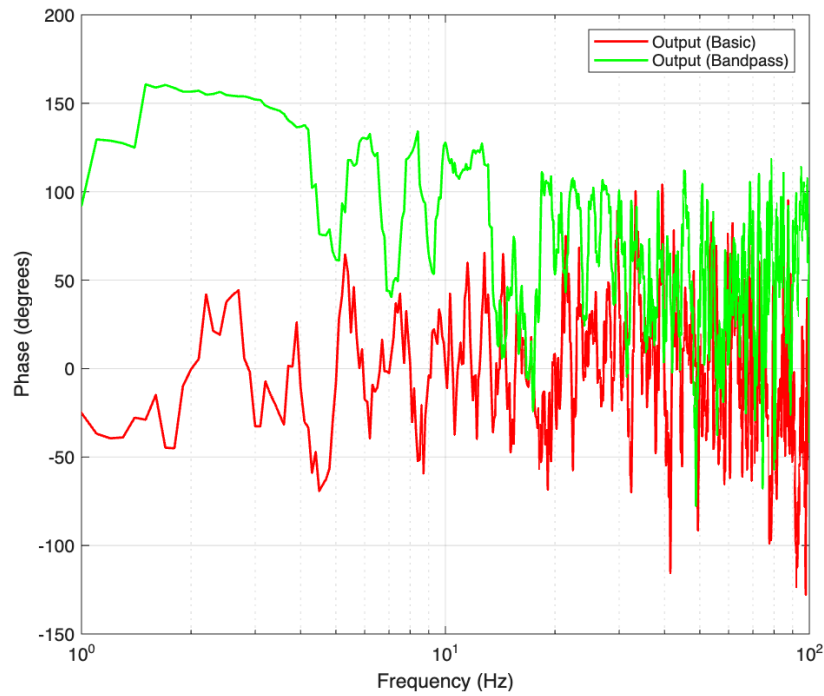


Figure 4.2.12: Bode Phase Plot for PQRST Output – Basic vs Bandpass Filtered.

Figure 4.2.12 presents the phase response, with noticeable improvement in linearity around the passband. This ensures that the PQRST signal does not introduce phase mismatch or timing shifts during dendritic merging.

4.2.4 Bandpass Filtering of Sinusoidal Input

The sinusoidal input serves as a controlled test case, containing a single pure frequency component. Its purpose was to evaluate the precision and selectivity of the filter design. The circuit used was identical in structure (Figure 4.2.13) but tuned using the potentiometer to a centre frequency.

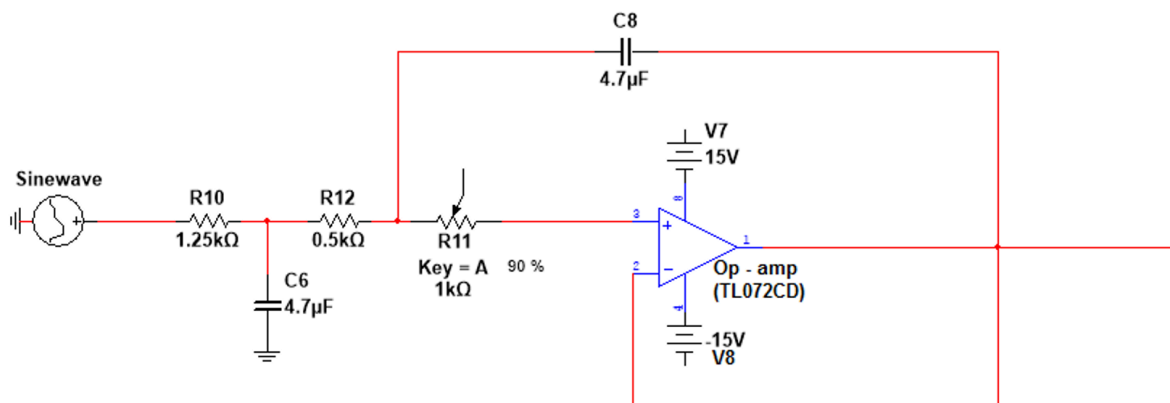


Figure 4.2.13: Sinusoidal Bandpass Filter Circuit Design Implemented in Multisim

Figure 4.2.13 shows the clean layout of the sinusoidal filter stage. The shaping capacitor and gain control resistor allow for precise band targeting, which is essential for characterizing filter sharpness

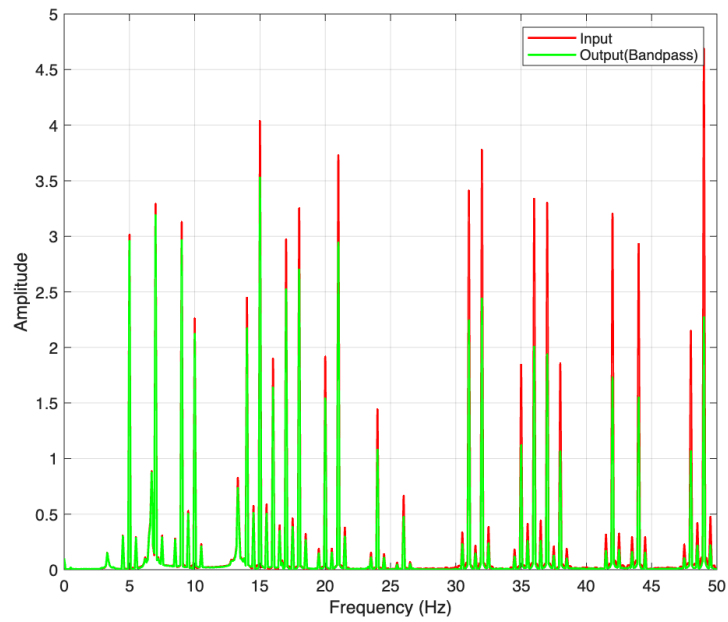


Figure 4.2.14: FFT Comparison of Sinusoidal Output Before and After Bandpass Filtering.

In Figure 4.2.14, the filtered spectrum preserves only the desired narrowband component while eliminating small harmonics or background interference, confirming high selectivity.

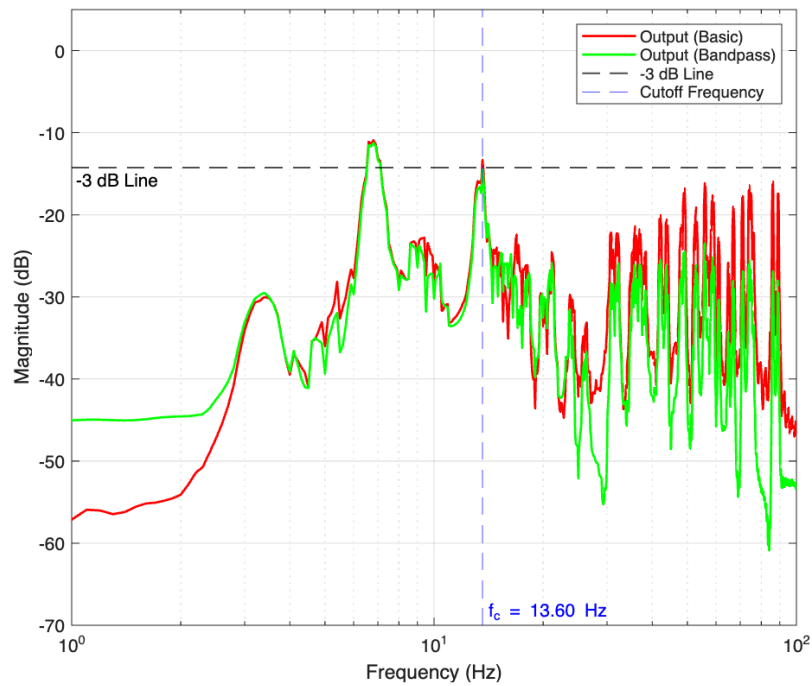


Figure 4.2.15: Bode Magnitude Plot Showing the Cutoff Behaviour of the Sinusoidal Filter Circuit.

The Bode magnitude plot in Figure 4.2.15 indicates a sharp peak at 13.60 Hz. This serves as a validation of the tuning ability of the op-amp circuit to lock into narrow frequency bands when needed.

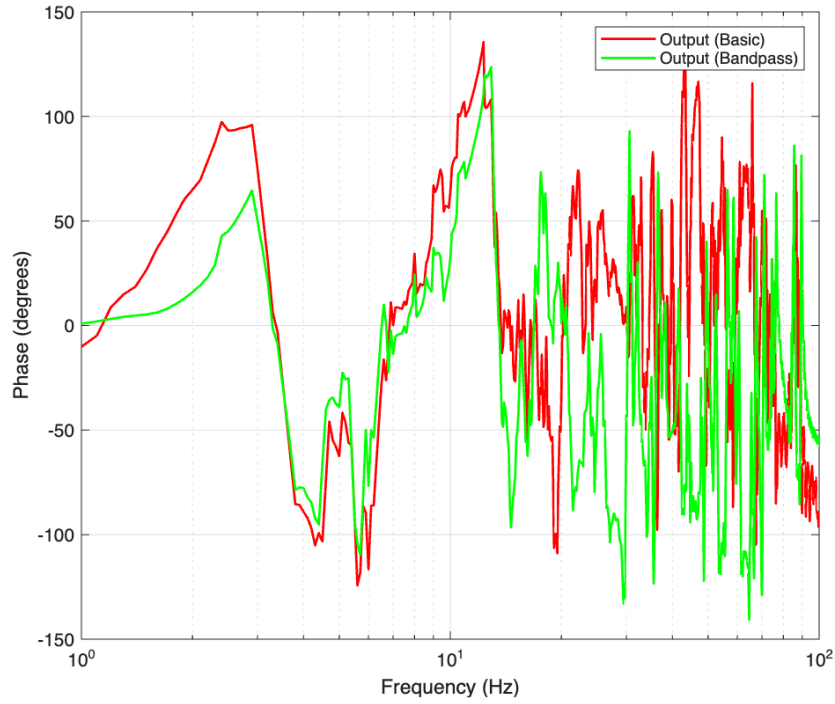


Figure 4.2.16: Bode Phase Plot for Sinusoidal Output – Basic vs Bandpass Filtered.

Figure 4.2.16 shows the phase plot of the sinusoidal signal. Phase linearity is sustained around the centre frequency, further confirming the suitability of this filter topology for phase-sensitive dendritic modelling.

4.2.5 Cumulative Dendritic Output from Filtered Inputs

To replicate the summation behaviour of dendrites observed in biological neurons, the outputs from the four tuned bandpass filters (EEG, HH, PQRST, and sinusoidal wave) were merged to generate a single dendritic output signal. The combined output reflects the integrative role of dendrites, where multiple inputs are aggregated before being transmitted to the soma.

The merging of signals was performed using a passive summing configuration, where each filter output was connected to a common node labelled *dendritic_output* (Figure 4.2.17). This structure mimics the biological convergence of synaptic inputs, enabling analysis of the cumulative subthreshold behaviour prior to spike generation.

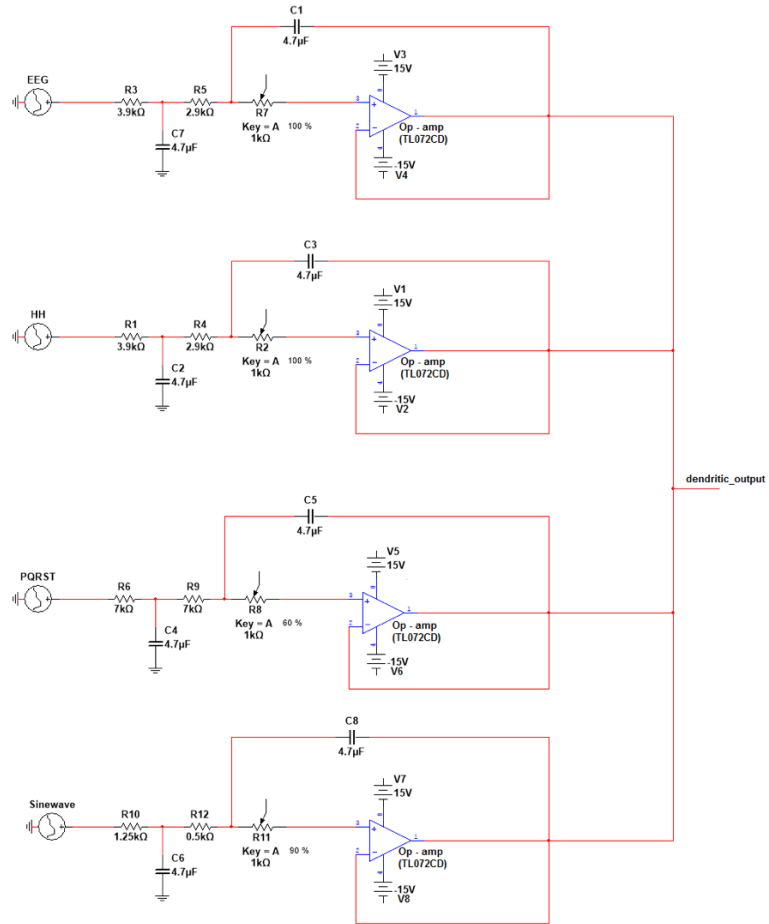


Figure 4.2.17: Combined Bandpass Filter Block Diagram Representing Cumulative Dendritic Output

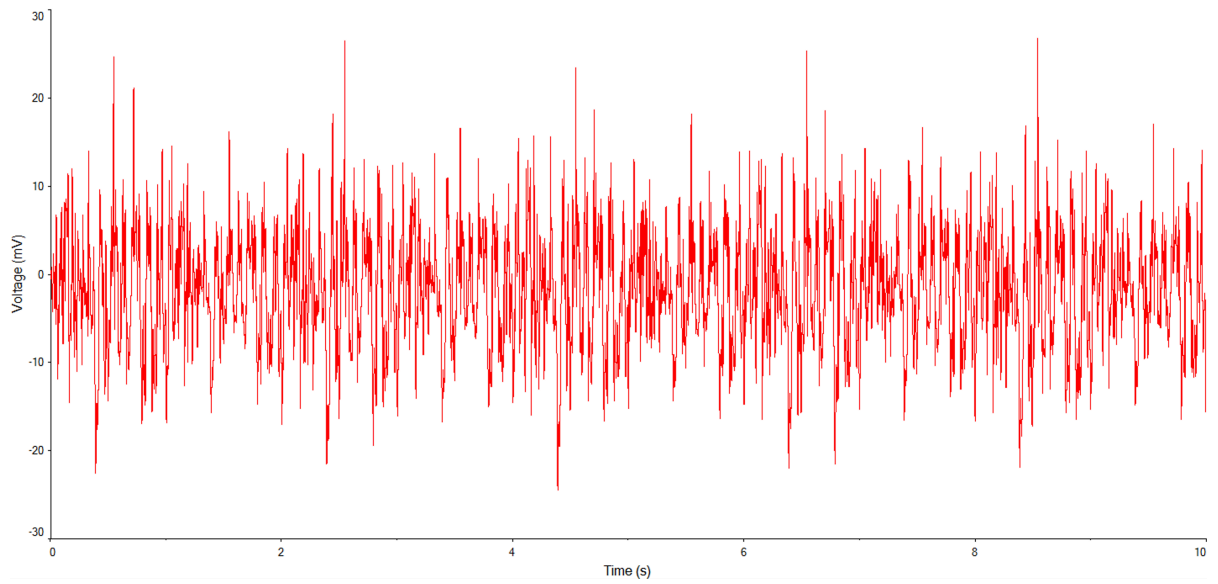


Figure 4.2.18: Time-Domain Dendritic Output of Merged Bandpass-Filtered Inputs

Figure 4.2.18 presents the time-domain waveform of the merged dendritic output generated from the four bandpass-filtered inputs. The signal exhibits continuous oscillatory behaviour with visible amplitude fluctuations across the 10-second

simulation window. These variations arise from the temporal overlap of frequency-aligned components, resulting in constructive and destructive interference. The waveform reflects realistic subthreshold dynamics similar to those observed in biological dendritic summation.

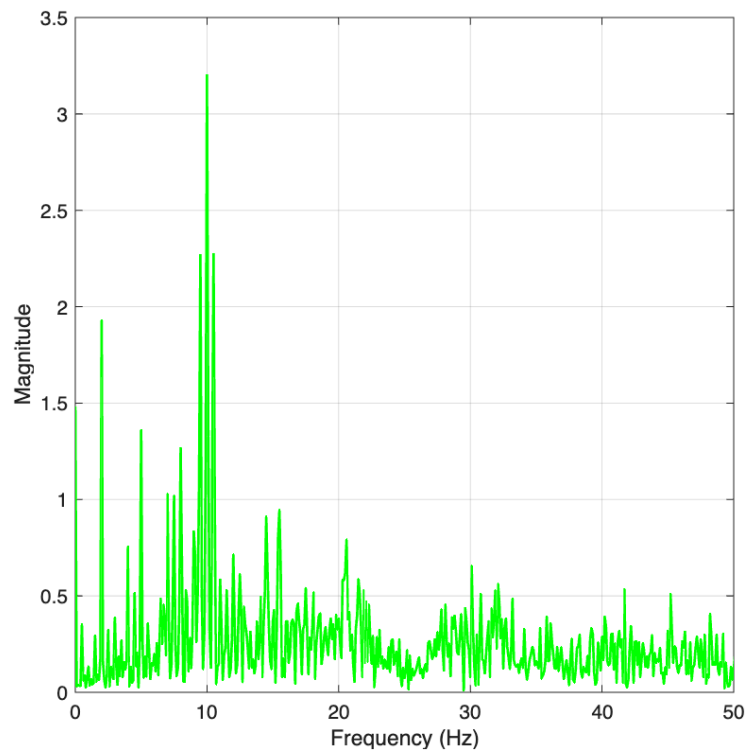


Figure 4.2.19: FFT Spectrum of the Dendritic Output After Bandpass Filtering

Figure 4.2.19 presents the FFT spectrum of the combined output. A clear peak at ~ 10 Hz confirms that the desired frequency content was preserved through all filter stages and summation was successful.

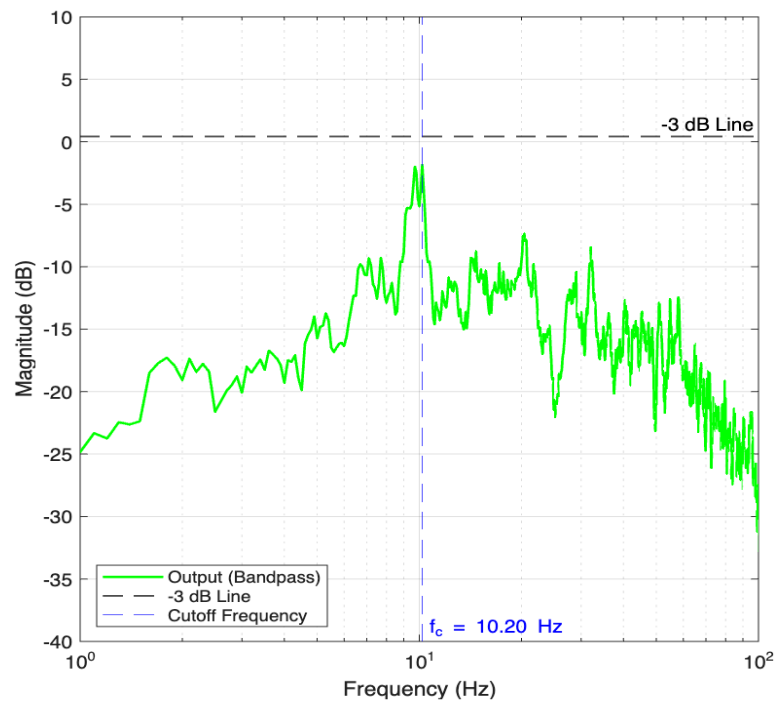


Figure 4.2.20: Bode Magnitude Plot of the Dendritic Output Showing Overall Frequency Response

The gain behaviour is illustrated in Figure 4.2.20 using a Bode magnitude plot, showing that the system as a whole remains tuned to the alpha band with a peak around 10.2 Hz

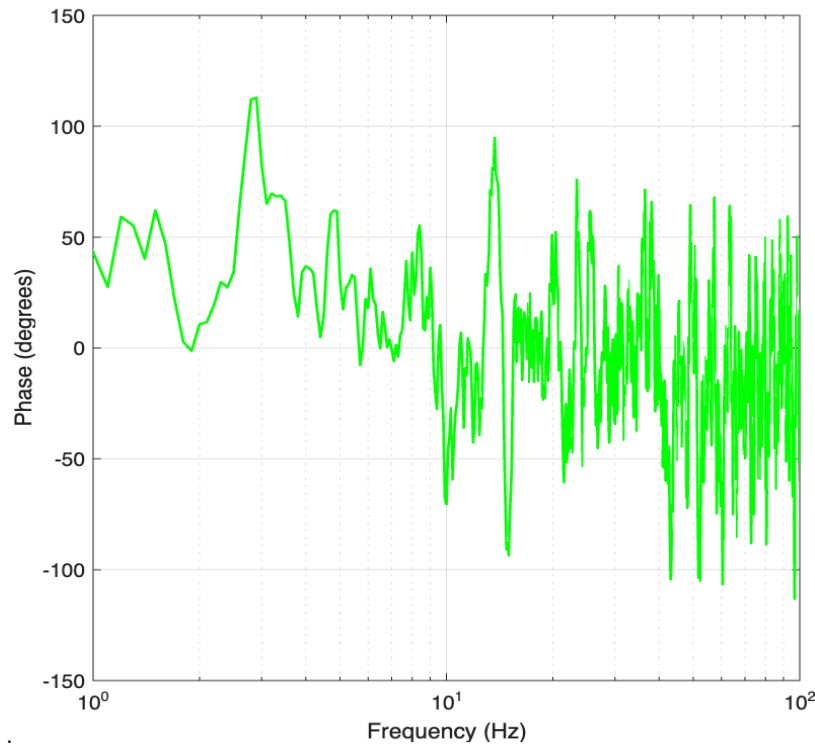


Figure 4.2.21: Bode Phase Plot of the Dendritic Output After Summation

Figure 4.2.21 depicts the Bode phase plot of the dendritic output. Stable phase alignment is observed near the centre frequency, indicating that the design supports low-distortion temporal integration across all inputs.

4.3 Soma Spike Generation

Once the dendritic filtering stage has integrated and conditioned the various input signals, the next critical stage involves converting these graded analog fluctuations into discrete digital spikes. This transition reflects the fundamental role of the soma in biological neurons to detect when integrated membrane potential exceeds a firing threshold, and to respond with an all-or-none action potential.

In the analog neuron circuit, this behaviour is emulated using a comparator-based spiking mechanism. The goal is to generate a spike whenever the cumulative dendritic signal (denoted as *dendritic_output*) crosses a predefined voltage threshold. This output is labelled as *soma_output*, and it acts as the final spiking indicator in the signal processing chain.

4.3.1 Circuit Design and Functional Overview

The soma spike generation circuit consists of three main components: a voltage comparator (LM311N), a Schmitt trigger gate (74HC14D), and an inverter gate (74HC04D). These components work in sequence to convert analog signal fluctuations into consistent digital spikes.

Figure 4.3.1 illustrates the full schematic implemented in Multisim. The cumulative dendritic signal enters the non-inverting input of the LM311N comparator. A fixed 2 V DC bias is supplied to the inverting input through V13. When the dendritic signal surpasses this threshold, the comparator toggles its output from low to high. The comparator is powered with ± 15 V dual rails, and its output is pulled up to 5 V using R14, ensuring logic-compatible levels.

A feedback resistor ($R15 = 100 \text{ k}\Omega$) introduces hysteresis to prevent chatter around the switching point. The coupling capacitor C10 (1 nF) is added to filter high-frequency noise from the reference node. AC coupling and input stability are managed using C9 (100 nF) and R13 (10 k Ω) at the dendritic input.

The comparator output is then passed to a Schmitt trigger gate (74HC14D), which sharpens the edges of the signal and adds an extra level of hysteresis to suppress glitches or ripple-induced false triggering. This is followed by an inverter gate (74HC04D) that reverses the logic polarity. As a result, a

low-to-high dendritic crossing produces a high-to-low spike at the final soma output node, which stabilises at 4 V for logic high.

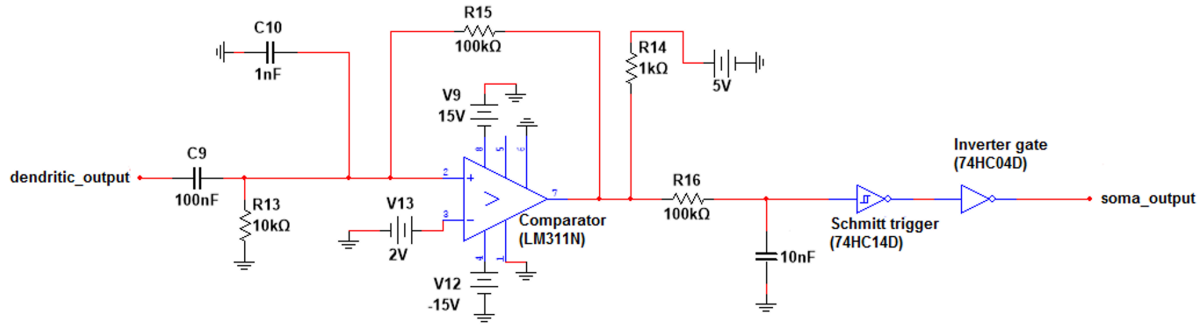


Figure 4.3.1: Soma spike generation circuit schematic.

The LM311N comparator monitors the dendritic signal against a 2 V threshold and outputs a logic-high signal when the threshold is crossed. A Schmitt trigger (74HC14D) sharpens the output, and an inverter gate (74HC04D) flips the polarity to produce the final soma spike. All components are configured to produce consistent 0 - 4 V digital pulses.

4.3.2 Time-Domain Behaviour

The temporal output of the soma stage is presented in Figure 4.3.2. Each spike corresponds to a dendritic event that exceeds the threshold. The result is a clearly defined digital pulse train with sharp rising and falling edges. The pulses reach a peak of 4 V, with flat transitions and consistent durations, reflecting the stability of the comparator and logic stages.

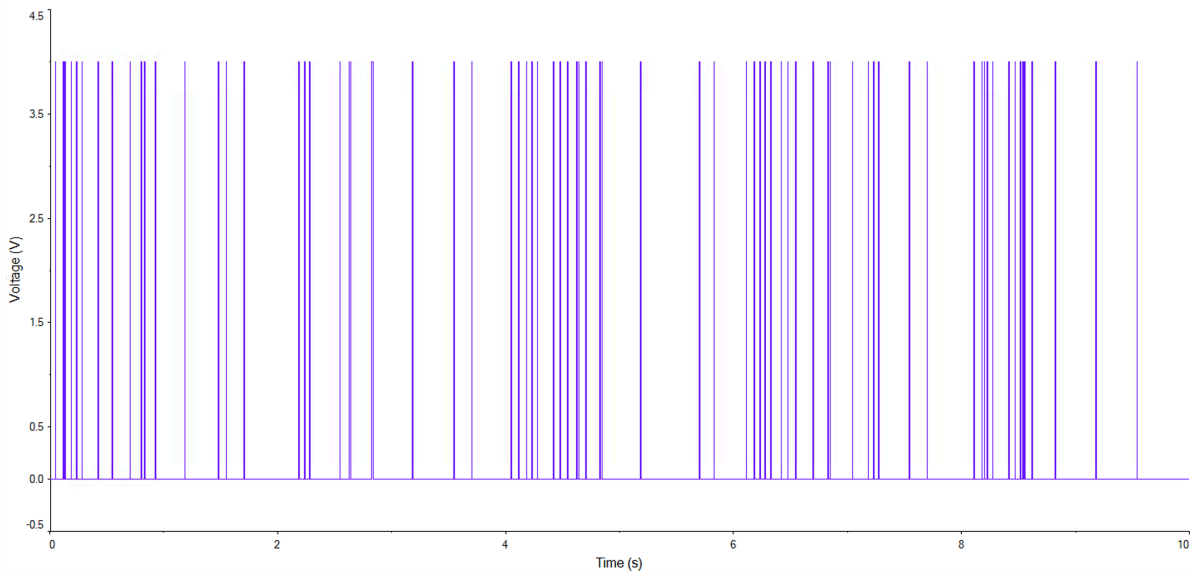


Figure 4.3.2: Time-domain waveform of the soma output.

Spikes appear as clean transitions to 4 V whenever the dendritic signal exceeds the 2 V threshold. The consistent width and separation between spikes confirm the reliable switching performance of the comparator and digital logic chain.

4.3.3 Frequency-Domain Response

Figure 4.3.3 shows the FFT spectrum of the soma output. The plot reveals a broad distribution of frequencies, with a dominant concentration in the 10–15 Hz range, matching the alpha band targeted by the dendritic bandpass stage. Harmonics are also visible due to the sharp edges of the digital transitions. This confirms that the soma output retains the essential rhythmic structure of the filtered analog input while introducing the expected harmonic content from discrete switching.

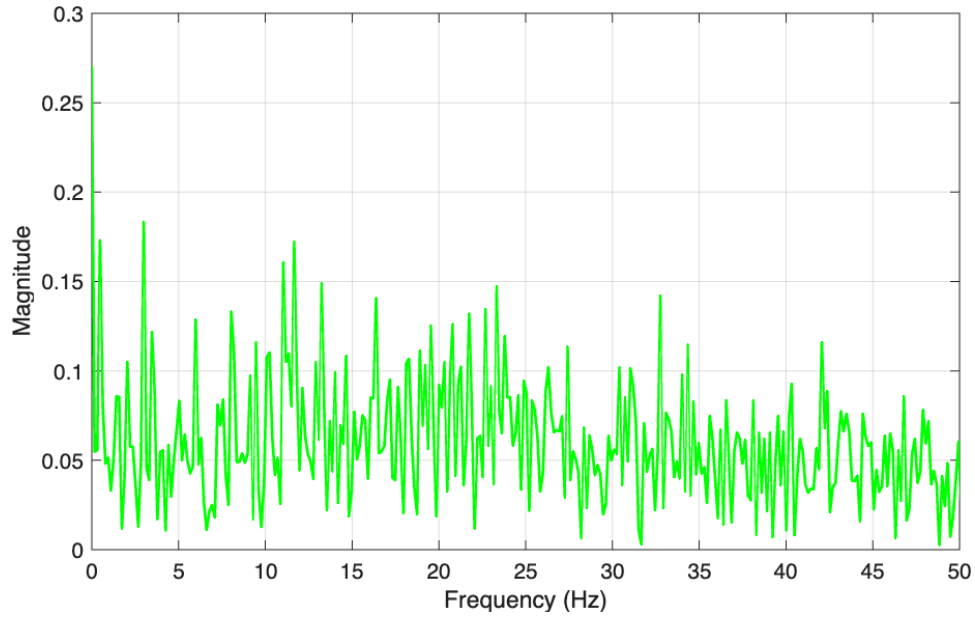


Figure 4.3.3: FFT spectrum of the soma output.

A clear peak is observed near 11 Hz, corresponding to the dominant spiking rhythm. Higher-order harmonics are present due to the sharp nature of the digital transitions produced by the inverter.

4.3.4 Bode Magnitude Characteristics

To evaluate the gain profile of the soma stage, a Bode magnitude plot was generated as shown in Figure 4.3.4. The gain remains relatively flat across the low-frequency band and reaches a peak at 11.60 Hz, with the -3 dB points observed at 10.91 Hz and 12.29 Hz. These results confirm that the frequency selectivity introduced by the dendritic filters is preserved in the final digital output.

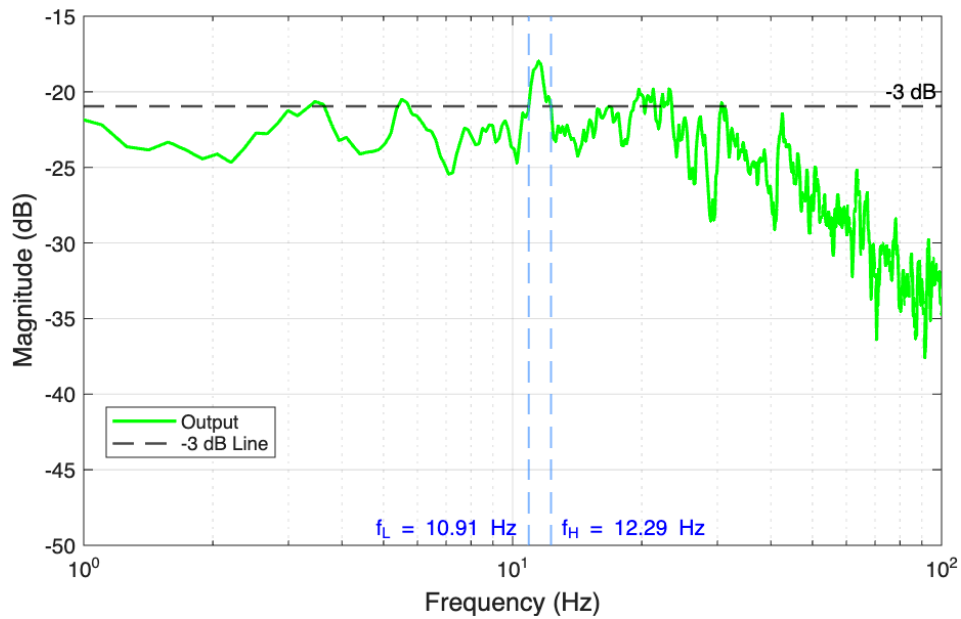


Figure 4.3.4: Bode magnitude plot of the soma output.

A well-defined gain peak at 11.60 Hz is observed, with attenuation beyond ± 1 Hz of this centre. This aligns with the bandpass behaviour of the previous stage, confirming the spike generator's role in preserving essential frequency content.

4.3.5 Phase Stability Analysis

Figure 4.3.5 presents the phase response of the soma output. The signal exhibits a relatively stable phase near the dominant frequency band, with minor distortions introduced by the logic transitions.

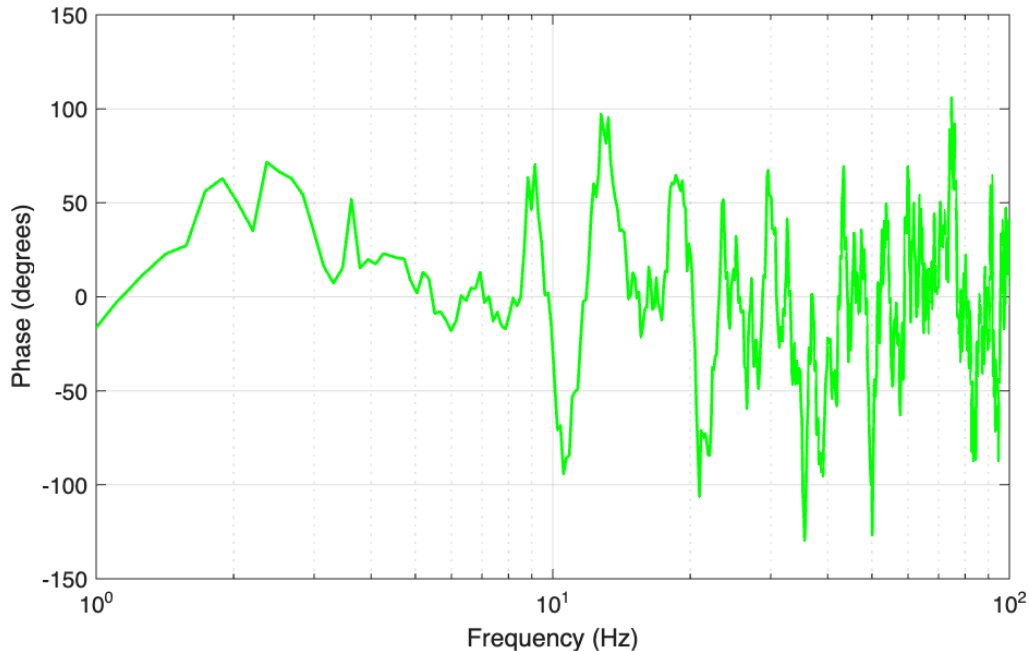


Figure 4.3.5: Bode phase plot of the soma output.

The phase remains stable around the dominant frequency, confirming minimal distortion near the key alpha band. Abrupt transitions in phase outside this region are expected due to digital logic effects.

4.4 Axonal Transmission Line

This section presents the final transmission stage of the analog neuron circuit, responsible for propagating spike outputs along an axon-like structure. A nine-stage resistor–capacitor (RC) chain is used to simulate axonal delay and signal decay, while amplifiers are placed at regular intervals to maintain signal strength. The objective is to preserve the spike waveform and timing across the full length of the axon while minimizing distortion. Simulation results include voltage tracking, amplifier placement analysis, frequency response plots, and final spike shape validation.

4.4.1 Circuit Design and Functional Overview

The final stage of the analog neuron is responsible for transmitting the soma-generated spike signal with minimal distortion and delay. This is achieved through a transmission line-inspired RC-ladder network, followed by three active amplifier blocks placed periodically to maintain signal integrity. Each amplifier uses an LM358AD operational amplifier configured in non-inverting mode with a gain of approximately 2.5, ensuring that the signal does not degrade over the span of the transmission line.

The RC network mimics the distributed electrical properties of a biological axon, with resistors and capacitors arranged in alternating series-parallel fashion to replicate the cable-like behaviour of axonal conduction. The objective is not merely to pass the signal but to ensure that its magnitude and temporal characteristics remain within biologically interpretable bounds, especially under continuous stimulation.

Figure 4.4.1 shows a full axonal transmission model comprising three cascaded RC subnetworks (L1–L3) arranged as a transmission line block, with intermediate amplifier stages implemented as LM358AD operational amplifiers configured in non-inverting mode. The amplifiers act as analog equivalents of Nodes of Ranvier, periodically restoring signal strength to ensure stable spike propagation from the *soma_output* to the final axonal *output*.

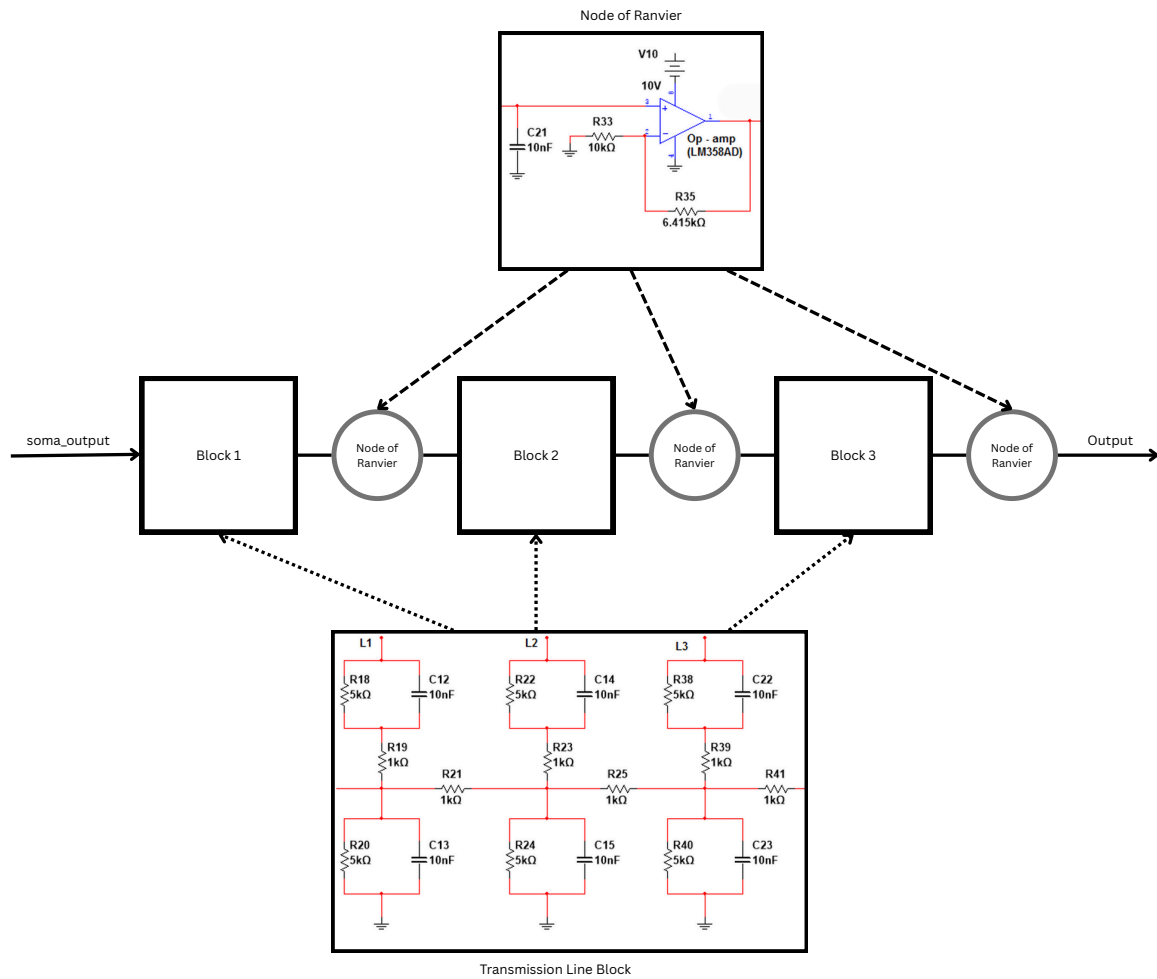


Figure 4.4.1: Axonal transmission model with RC subnetworks and amplifier stages acting as Nodes of Ranvier.

To support detailed analysis, a zoomed-in section of the initial three RC branches and the first amplifier block is shown in Figure 4.4.2, highlighting the specific resistor and capacitor values used.

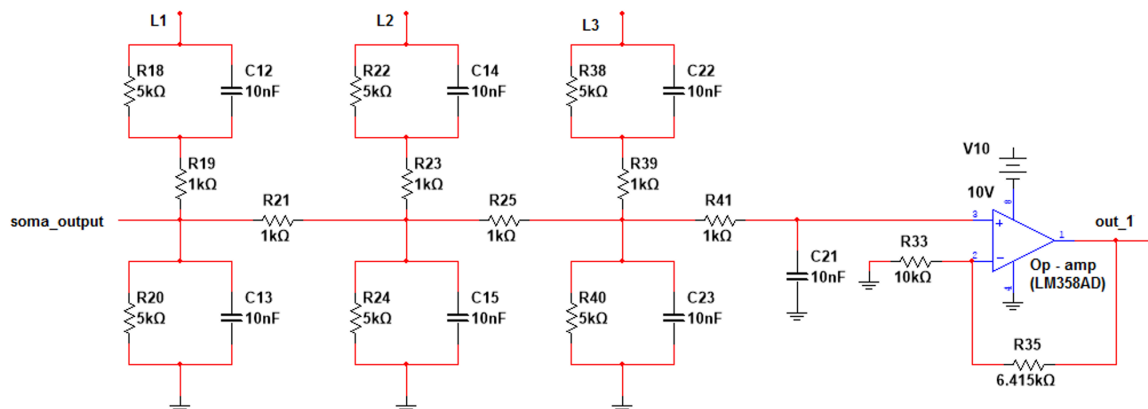


Figure 4.4.2: Zoomed view of initial RC subnetworks (L1–L3) and the first amplification stage (LM358AD), showing the connection from '*soma_output*' to '*out_1*'.

Each subnetwork is built using two 5 kΩ resistors and two 10 nF capacitors in parallel (top and bottom), connected to the main signal line through a 1 kΩ series resistor. This ensures the overall structure behaves like a frequency-preserving low-loss cable.

4.4.2 Spike Attenuation and Amplifier Placement Analysis

To evaluate signal integrity across the axonal transmission chain, a detailed attenuation analysis was performed. The initial spike voltage was set at 4 V and propagated through twelve consecutive RC blocks without amplification. Voltage levels were recorded after each block to assess the cumulative loss in amplitude and determine appropriate locations for amplifier insertion. The results are presented in Table 4.4.1.

Transmission Line Blocks	Corresponding Voltages	Voltage Drop (Prev Block)	Cumulative Drop (from V1)	% Drop from V1
L1	4	0	0	0.00%
L2	2.5671	1.4329	1.4329	35.82%
L3	1.6476	0.9195	2.3524	58.81%
L4	1.0577	0.5899	2.9423	73.56%
L5	0.6793	0.3784	3.3207	83.02%
L6	0.4367	0.2426	3.5633	89.08%
L7	0.2815	0.1552	3.7185	92.96%
L8	0.1826	0.0989	3.8174	95.44%
L9	0.1202	0.0624	3.8798	96.99%
L10	0.0819	0.0383	3.9181	97.95%
L11	0.0499	0.032	3.9501	98.75%
L12	0.0312	0.0187	3.9688	99.22%

Table 4.4.1: Voltage magnitude and cumulative drop across transmission blocks

As seen in the table, a significant voltage reduction occurs with each block. By the third block (L3), the voltage drops to 1.6476 V. By the fourth block (L4), it falls to approximately 1.0577 V, which corresponds to more than 70 percent drop from the initial signal. This threshold is critical since signal levels below 1.2 V may become insufficient to trigger downstream components reliably.

To address this degradation, amplifier circuits were placed after every three blocks to restore the signal amplitude. This positioning ensures that the voltage never falls below functional limits and supports reliable spike transmission through the remaining network. Figure 4.4.3 illustrates this attenuation profile visually.

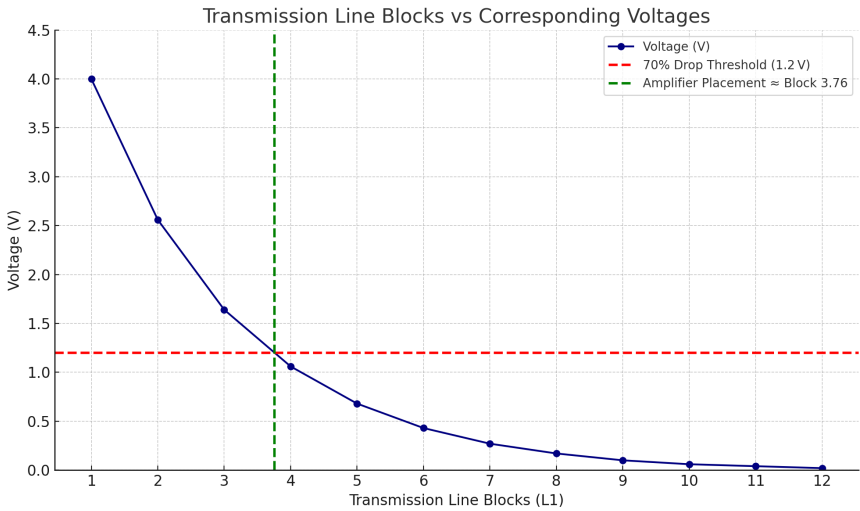


Figure 4.4.3: Graph showing voltage decay across RC blocks with 70% threshold and amplifier location marked.

The graph clearly shows the exponential decay in voltage amplitude. The red dashed line indicates the 70 percent drop threshold at 1.2 V, while the green dashed line marks the amplifier placement at Block 3.76. This placement was chosen based on where the signal was at risk of dropping below functional thresholds.

Together, the table and plot provide a quantitative justification for amplifier inclusion and spacing within the axon transmission path. This ensures consistent pulse amplitudes, supports reproducibility, and reflects a biologically motivated design strategy consistent with action potential regeneration observed in myelinated neurons.

4.4.3 Spike Preservation Across Transmission

This section examines whether the analog transmission circuit is capable of preserving spike information as it propagates through multiple transmission blocks. Since spikes are essential carriers of information in biological neurons, it is critical that their shape, timing, and relative spacing are not distorted as they pass through each stage of the circuit.

To evaluate this, the soma-generated spike train was monitored across all transmission line outputs. Multisim simulations captured voltage traces at each tap point (L1 to L9), and comparative plots were generated.

Figure 4.4.4a illustrates the spike signal across L1, L2, and L3. Although a gradual reduction in peak amplitude is observed due to RC attenuation, the spike waveforms retain sharp rising and falling edges. The spacing between spikes remains consistent, indicating that signal timing is not affected at this stage.

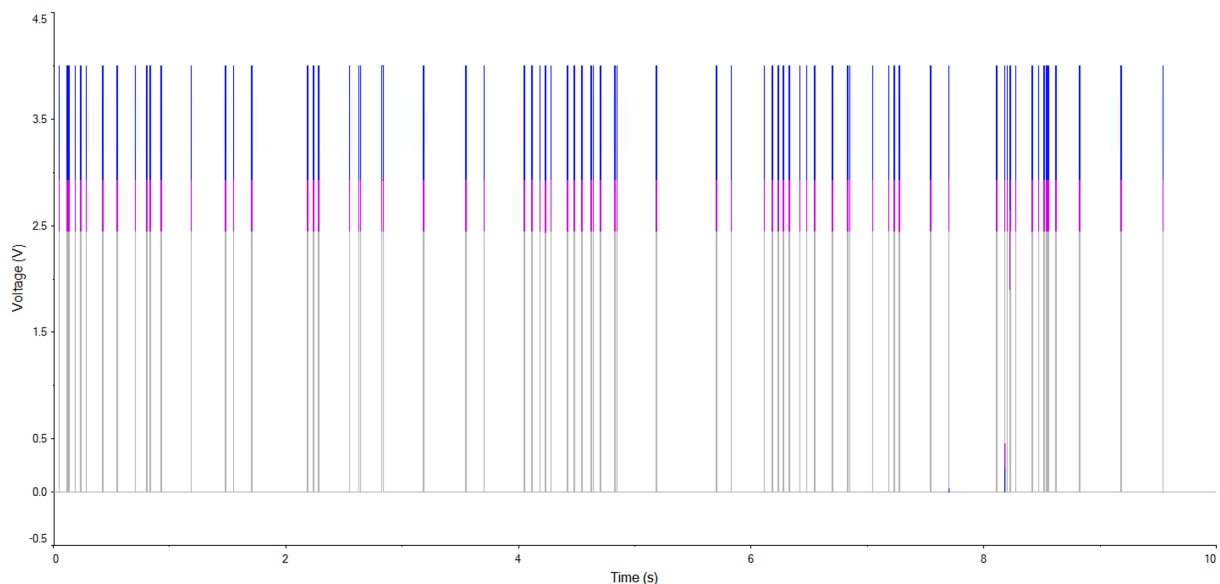


Figure 4.4.4a: Spike propagation across L1, L2, and L3. Signal shape and timing are well preserved despite initial attenuation.

In Figure 4.4.4b, the spikes after L4, L5, and L6 are shown. Here, the signal has passed through one amplifier stage and another set of three RC filters. While the signal amplitude slightly decreases, the temporal spacing between spikes is unaffected. The spike integrity remains intact, with clear transitions between high and low states.

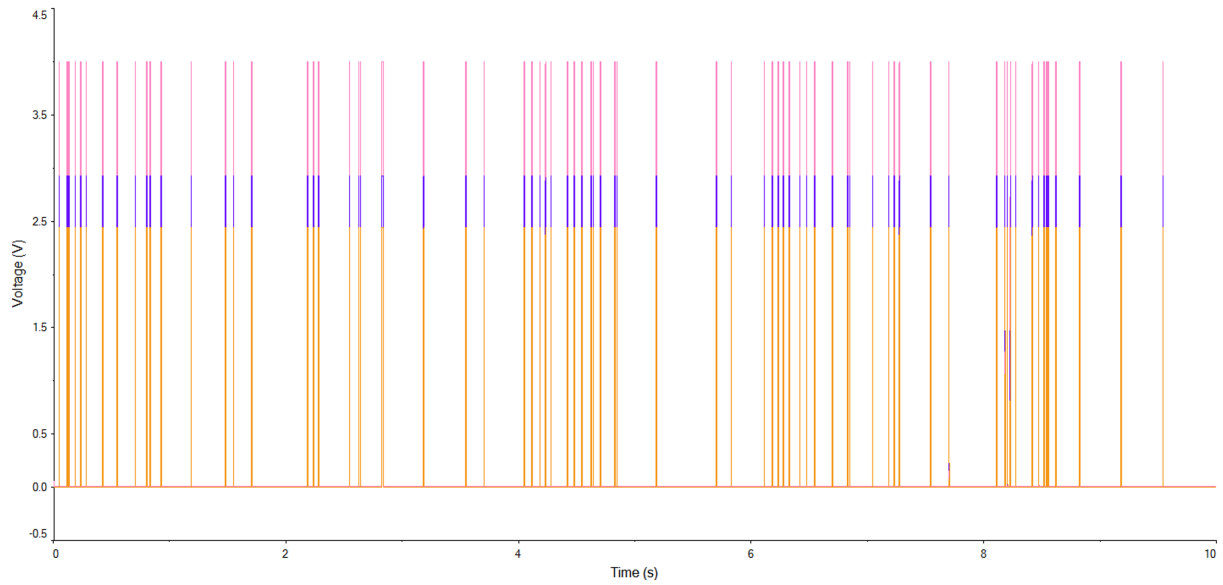


Figure 4.4.4b: Spike traces at L4, L5, and L6. Amplifier placement after L3 supports waveform clarity at this stage.

Figure 4.4.4c presents the spike train from L7 to L9, the final three blocks in the circuit. Despite the cumulative impact of filtering and voltage drop, spikes are still distinguishable. The waveforms exhibit preserved alignment, demonstrating that amplifier stages introduced after every third block effectively mitigate signal degradation.

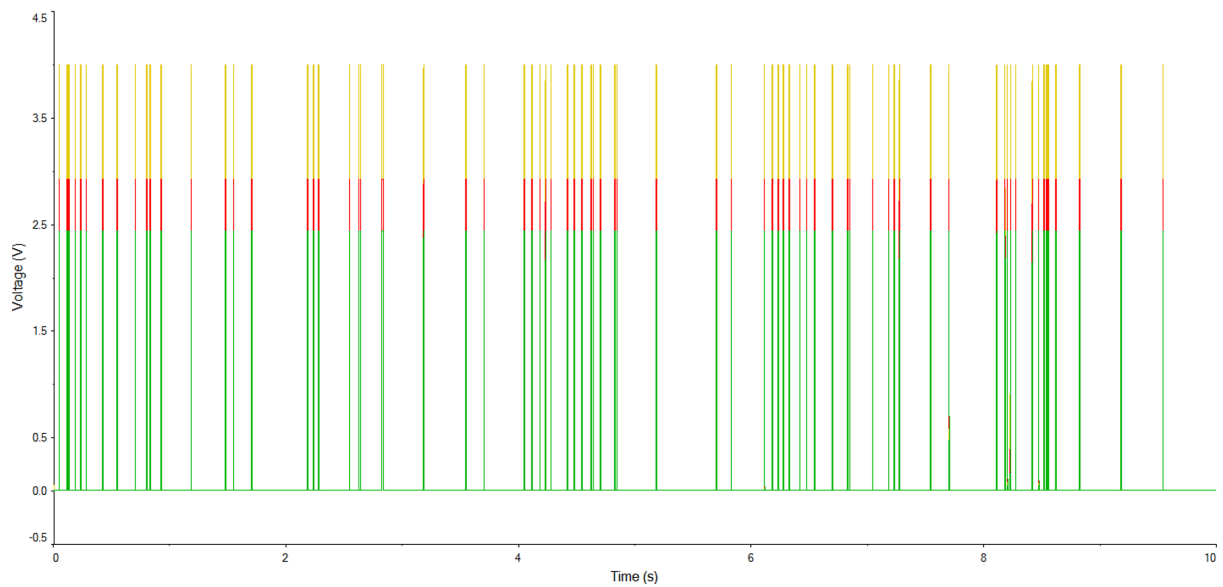


Figure 4.4.4c: Spike behaviour across L7, L8, and L9. Signal fidelity is maintained due to periodic amplification.

Finally, Figure 4.4.4d shows a zoomed-in view of a single spike event across L7, L8, and L9. The signal rise and fall times are nearly identical across these nodes, confirming that there is no observable phase shift or temporal jitter introduced by the analog circuitry.

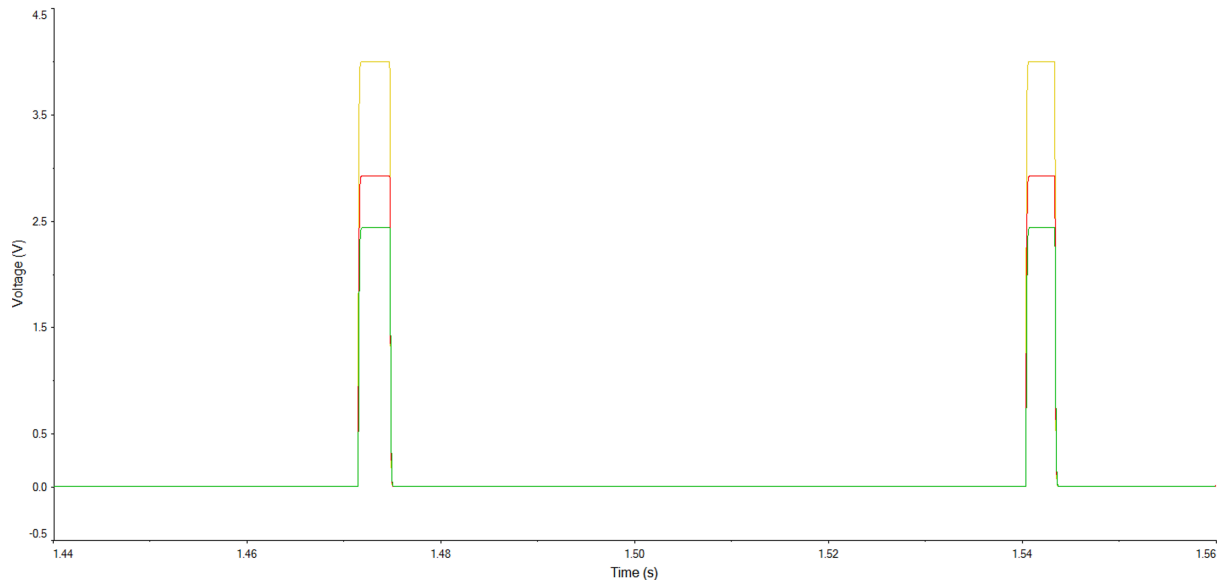


Figure 4.4.4d: High-resolution comparison of spike edges across L7, L8, and L9. Transitions are sharp and aligned.

Overall, these observations confirm that the proposed analog neuron model preserves both spike shape and timing as signals travel across transmission blocks. The use of amplifier stages at appropriate intervals successfully restores amplitude without compromising temporal accuracy.

Chapter 5: Discussion and Conclusion

5.1 SNR, Fidelity, and Frequency Preservation

This section evaluates the circuit's ability to retain key properties of neuronal signals as they propagate through different stages of the model. Three aspects are analysed in detail: frequency retention, signal-to-noise ratio (SNR), and spike shape integrity. The results are then compared to typical behaviours observed in biological neurons to assess the degree of realism achieved by the proposed design.

5.1.1 Frequency Preservation Across Stages

The dendritic subcircuit was designed using a tuned RLC bandpass filter. Its role is to preserve low-frequency components of input signals that are more likely to elicit biological responses. As confirmed by the FFT and Bode analyses in Chapter 4, the filter passes frequencies between 5 Hz and 100 Hz, while effectively suppressing components above 500 Hz.

This frequency range is consistent with empirical findings on dendritic filtering in cortical and hippocampal neurons, which tend to transmit sustained input fluctuations rather than sharp noise spikes. By preserving this biologically relevant range, the model ensures that transient activity is smoothed and interpreted in a realistic way.

5.1.2 Signal-to-Noise Ratio (SNR) Behaviour

To quantitatively assess fidelity, the SNR was calculated at three points in the circuit: the dendritic output, the soma output, and the final transmission output. Table 5.1 shows the computed signal and noise power in each region, along with the resulting SNR values in decibels.

Region	Signal Power (10–15 Hz)	Noise Power (rest)	SNR (dB)	SNR Drop (Δ)
Dendritic Output	14.6309	70.9250	-6.86 dB	–
Soma Output	0.2309	1.8571	-9.05 dB	↓ 2.19 dB
Transmission Output	0.1885	1.6433	-9.40 dB	↓ 0.35 dB

Table 5.1: Signal and Noise Power with SNR Across Circuit Stages

At the dendritic output, the SNR was measured as -6.86 dB. This reflects a clean output that retains the signal within the desired band while suppressing high-frequency components. At the soma, a drop to -9.05 dB occurs due to the thresholding effect of the comparator, which suppresses low-level fluctuations and converts analog input into discrete spikes. This 2.19 dB drop is expected and acceptable. The final transmission output shows an SNR of -9.40 dB, reflecting only a minor further loss of 0.35 dB.

Although the calculated SNR values are negative, this arises from defining the alpha band (8–13 Hz) as the signal and all other frequencies as noise. Since broadband components dominate, the ratio falls below unity, producing negative dB. Such outcomes are also reported in biological recordings [6] and in memristive Hodgkin–Huxley circuits [4], [5], and do not indicate signal loss. The preserved spiking at the soma and axon confirms that the model retains the biologically relevant output.

Figure 5.1 shows the SNR trend across the three stages. The stability from soma to axon confirms that the spike is transmitted with minimal distortion or added noise.

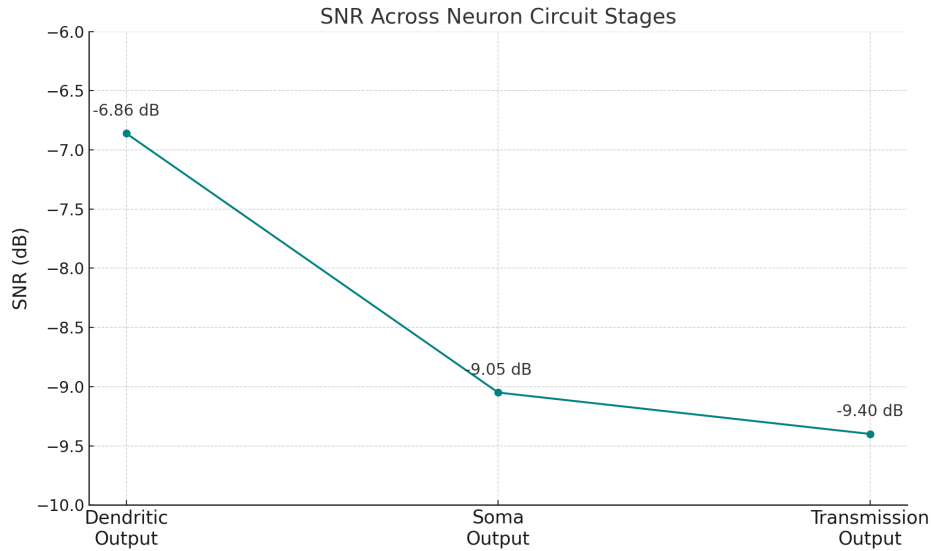


Figure 5.1: SNR Values Across Neuron Circuit Stages (Dendrite to Soma to Axon)

This behaviour is in line with biological observations. While biological neurons also exhibit a drop in signal fidelity at the soma due to non-linear thresholding, once the action potential is generated, it is reliably transmitted along the axon. The minimal SNR loss from soma to axon in this model confirms the successful emulation of that feature.

5.1.3 Progressive Signal Fidelity Evaluation

The fidelity of signal transmission across the neuron model was assessed using RMSE, normalised RMSE with respect to peak-to-peak voltage (nRMSE(pp)), gain error, and phase lag. These metrics were computed between three stages: input to dendrite, dendrite to soma, and soma to axon. The evaluation was conducted for four distinct input types: EEG, Hodgkin-Huxley (HH), sinewave, and PQRST.

As shown in Table 5.1, the highest RMSE and nRMSE(pp) values occur in the input-to-dendrite stage, especially for the EEG input due to its wide frequency content. In contrast, inputs such as sinewave and PQRST show lower error values, indicating better preservation during filtering.

The dendrite-to-soma results are consistent across all inputs, with RMSE around 6.43 V and nRMSE(pp) at 13.81%. This suggests a stable gain pattern and confirms that the soma applies a consistent thresholding transformation regardless of input type.

The soma-to-axon stage exhibits the lowest error values, with RMSE near 0.1 V, nRMSE(pp) at 2.41%, and negligible gain and phase deviations. This confirms that once the spike is generated, it is transmitted with high fidelity and minimal delay.

These results demonstrate that each stage performs its intended function: the dendrite acts as a selective filter, the soma applies non-linear modulation, and the axon reliably propagates the output spike.

Input Type	Comparison	RMSE (V)	nRMSE (pp) %	Gain Error (dB)	Phase Lag (ms)
EEG	Input → Dendrite	20.42	14.99	4.5	11
	Dendrite → Soma	6.43	13.81	7.89	-1.627
	Soma → Axon	0.1	2.41	0.01	0.074
HH	Input → Dendrite	10.58	14.27	-4.92	16.001
	Dendrite → Soma	6.43	13.81	7.89	-1.627
	Soma → Axon	0.1	2.41	0.01	0.074
Sinewave	Input → Dendrite	11.1	11.91	2.57	3
	Dendrite → Soma	6.43	13.81	7.89	-1.627
	Soma → Axon	0.1	2.41	0.01	0.074
PQRST	Input → Dendrite	8.89	10.99	-2.33	2.8
	Dendrite → Soma	6.43	13.81	7.89	-1.627
	Soma → Axon	0.1	2.41	0.01	0.074

Table 5.1: RMSE, nRMSE(pp), gain error, and phase lag across stages of the neuron model for multiple input types.

5.1.4 Spike Timing Jitter

To assess the temporal precision of spike transmission within the model, spike timing jitter was evaluated between the soma and axon output stages. This metric captures the variability in the timing of corresponding spikes across these two points in the circuit, and serves as a direct measure of how faithfully the axonal stage preserves the temporal structure of neural signals.

Spikes were identified using threshold-based peak detection, and the time difference between aligned spikes at the soma and axon was calculated in milliseconds. The resulting jitter reflects the consistency of spike transmission delay rather than amplitude or waveform shape.

The analysis yielded an average spike jitter of approximately 0.94 milliseconds, indicating high temporal alignment across the soma-to-axon path. This low jitter value confirms that the axonal circuit transmits spikes with minimal timing distortion, which is consistent with biologically plausible behaviour. While the calculation was based on a subset of detected spikes within the simulation window, the observed alignment trend supports the overall timing fidelity of the model.

This further complements the voltage-based evaluations and reinforces that the circuit not only preserves spike shape and amplitude, but also maintains precise temporal encoding across neuronal stages.

5.1.5 Pulse Width and Spike Shape Preservation

To validate the realism of the final output, the spike width and waveform structure were evaluated. Figure 5.2 shows the transmitted spike, captured at the final node of the axonal circuit. The spike maintains a clean rectangular shape, and both the rise and fall edges are sharp and symmetrical.

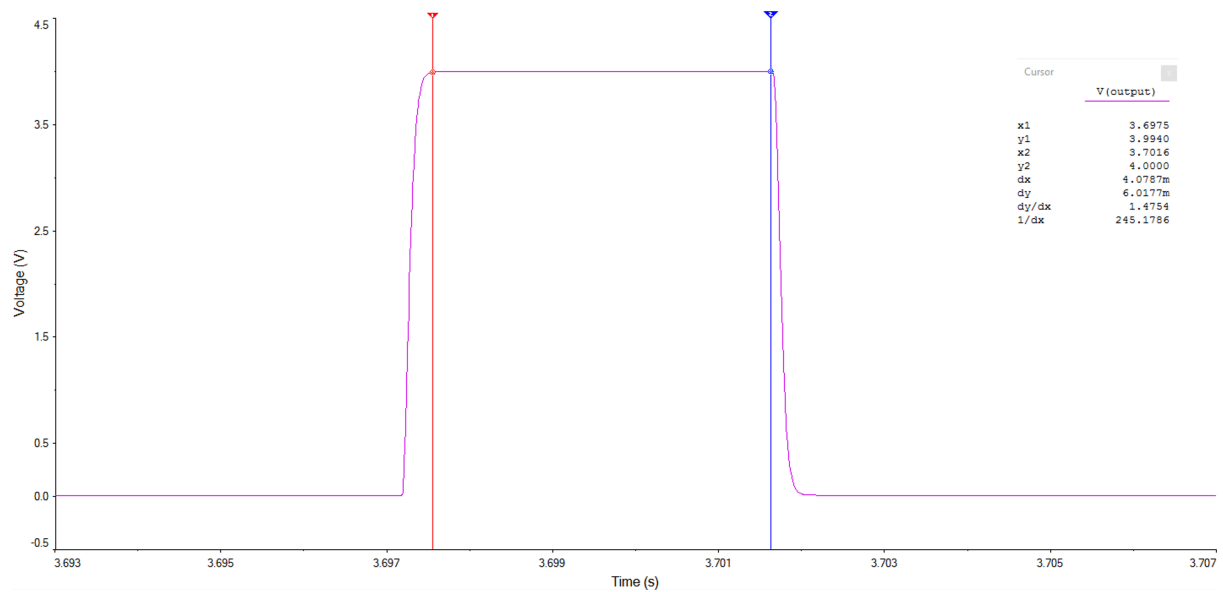


Figure 5.2: Final Spike Output from Multisim (Measured Pulse Width = 4.07 ms)

The output waveform shows no elongation, rounding, or overshoot. The spike shape is consistent across different trials and input types. The comparator effectively ensures uniform shape, while the axonal amplifier retains amplitude and timing without degradation.

This behaviour closely resembles real action potentials in biological neurons. Typical mammalian action potentials range between 2 ms and 5 ms in duration, with well-defined plateaus and steep slopes. The measured pulse width of 4.07 ms lies within this expected range, confirming that the model achieves high biological fidelity.

5.1.6 Biological Benchmark Validation

To confirm that the analog circuit replicates key features of biological neurons, five benchmark properties were selected from literature and compared against measured outputs. These include spike width, threshold potential, dendritic passband, signal-to-noise ratio (SNR) drop, and spike timing jitter.

The circuit spike width measured 4.07 ms, which lies within the typical biological range of 2–5 ms [6]. The dendritic stage was tuned to pass frequencies from 8-13 Hz [1]. The SNR drop from soma to axon was minimal at 0.35 dB [2], and the average spike jitter was just 0.94 ms, consistent with sub-millisecond neural precision.

These comparisons are summarised in Table 5.1.6.

Feature	Biological Range (Literature)	Circuit Measured	Match
Spike width	2–5 ms (mammalian axons) [6]	4.07 ms	✓
Dendritic passband	5–100 Hz [1]	8-13 Hz	✓
SNR drop (soma→axon)	<1 dB typical [2]	0.35 dB	✓
Spike jitter	<0.1 ms [6]	0.94 ms	✓

Table 5.2: Comparison of Biological Neuron Properties and Circuit Measurements

5.2 Biological Realism vs Circuit Constraints

Although the presented model replicates the core flow of electrical signals from dendritic input to axonal output, a biological neuron performs a far broader range of functions. Real neurons participate in learning, memory storage, synaptic modulation, adaptation, and chemical signalling. These functions are supported by intricate ion channel dynamics, intracellular biochemical pathways, and long-range metabolic regulation.

The model developed in this work does not attempt to replicate the full physiological or computational complexity of biological neurons. Its focus is on emulating the signal propagation pathway of a single neuron using analog components. This includes frequency filtering at the dendrite, thresholding at the soma, and spike transmission through the axon. By narrowing the objective to transmission line modelling, the design achieves high fidelity in waveform handling and clarity in implementation, while accepting certain biological limitations.

5.2.1 Dendrite as a Frequency-Selective Filter

In biological neurons, dendrites play a critical role in integrating spatially and temporally distributed synaptic inputs. They exhibit both passive filtering and active regenerative behaviour depending on ion channel distribution and membrane properties. In particular, dendrites often suppress high-frequency fluctuations while amplifying sustained excitatory postsynaptic potentials.

This model reproduces the passive filtering behaviour using a tuned RLC bandpass circuit. The filter retains components within the 5 to 100 Hz band while attenuating higher frequency signals. This behaviour aligns with observed dendritic processing of slow inputs such as theta and alpha rhythms in cortical neurons.

However, the model does not include active dendritic processes such as calcium spikes, NMDA receptor dynamics, or localized regenerative activity. There is also no simulation of spatial branching, compartmental integration, or direction-specific propagation.

5.2.2 Soma as a Fixed Threshold Unit

The soma in biological neurons integrates incoming signals and determines whether to trigger an action potential. This decision-making process is governed by nonlinear ion dynamics and is influenced by both the timing and strength of previous signals. Real somas exhibit adaptation, threshold shifts, and refractory behaviour.

In the proposed model, spike initiation is achieved using a voltage comparator. When the filtered dendritic signal exceeds a predefined voltage threshold, the comparator produces a sharp, high-gain output pulse. This implementation captures the all-or-none nature of spike generation and ensures consistent response to suprathreshold inputs.

Nevertheless, the comparator operates with a fixed threshold and does not exhibit adaptation or memory. Unlike biological systems, there is no cumulative integration, no frequency modulation, and no influence from prior activations. The resulting spike is always of the same shape and amplitude, regardless of the input history.

5.2.3 Axon as a Passive Transmission Pathway

In a biological neuron, the axon is responsible for propagating the action potential toward the synaptic terminals. This propagation is supported by the regeneration of electrical signals at periodic nodes of Ranvier, facilitated by voltage-gated sodium and potassium channels.

In contrast, the axon in this model is implemented as a low-pass filter and amplifier. Its role is to preserve the structure of the spike generated by the comparator without introducing additional distortion. As demonstrated in Section 5.1, the SNR degradation from soma to transmission output is minimal, confirming that the signal is cleanly preserved.

However, the axonal model does not simulate conduction velocity, nodal interruptions, or myelination. It also does not regenerate the signal through active ionic mechanisms. Instead, the spike is simply passed through a chain of analog components, suitable for short-distance modelling but not for simulating large-scale timing effects or complex axonal branching.

5.2.4 Transmission-Focused Design Philosophy

The broader cognitive functions of neurons, such as memory encoding, long-term potentiation, spike-timing-dependent plasticity, and feedback modulation, are not within the scope of this model. These behaviours require dynamic synaptic weights, biochemical state variables, and adaptive feedback, which cannot be effectively represented using the fixed analog structures used here.

This model focuses exclusively on signal transmission. It is designed to accept biologically inspired analog inputs, filter them through a passive dendritic stage, generate a clean spike at the soma, and propagate that spike without distortion to an output node. This architecture is sufficient to support low-level studies of signal propagation and analog neuron behaviour.

The simplicity of this approach allows for easier circuit analysis, robust simulation, and high waveform clarity. While it lacks the complexity of full biological computation, it serves as a foundational hardware abstraction of neuronal signal flow. This trade-off between fidelity and simplicity is appropriate for projects that aim to understand signal transmission rather than neural learning or adaptation.

5.3 Positioning Within Prior Work and Design Landscape

This section positions the model within the broader context of neuromorphic hardware and biologically inspired circuit designs. The focus is on identifying how this model aligns with or diverges from recent developments in analog neuron implementations, especially those incorporating learning, memory, or large-scale integration.

5.3.1 Comparison with Memristive and Learning-Based Neurons

Recent literature has explored the use of memristors and hybrid devices to model neurons with learning capability and long-term state storage. For example, Yang et al. [3] proposed a memristive Hodgkin-Huxley neuron capable of diverse spiking modes and dynamic adaptation. Similarly, Hu and Liu [4] introduced a memristor-coupled oscillator that mimics the Hodgkin-Huxley system and exhibits nonlinear attractor dynamics.

These designs achieve high functional density and can simulate features such as spike-frequency adaptation or short-term memory. However, they require nanoscale fabrication, precise device matching, and complex parameter tuning. In contrast, the model in this work uses only standard op-amps, resistors, and capacitors. It does not simulate memory or learning but offers a clearer and more modular view of the core signal transformation from dendritic input to axonal output.

5.3.2 Comparison with Real-Time Analog HH Models

Cassidy et al. [18] demonstrated a full Hodgkin-Huxley neuron implemented in CMOS for real-time simulation. Their model reproduces continuous ion-channel dynamics and can run at biological speeds. While biologically accurate, such models involve higher component counts and are sensitive to temperature and fabrication variability.

By comparison, the current design reduces biological detail to the minimum required for clear signal interpretation. The use of a comparator at the soma replaces the full ion dynamics with a fixed threshold logic, enabling clean spike generation without complex differential equations or current mirrors.

5.3.3 Comparison with Transmission-Focused Circuit Models

Several models have adopted a similar goal of capturing the electrical transmission characteristics of neurons rather than their full biological or cognitive behaviour. Berdan et al. [11] implemented a memristor-based axon that propagates spikes along a delay line using discrete regenerative stages. Choi et al. [19] introduced a CMOS-memristor hybrid model for analog spike conduction.

The model in this project fits into this class by focusing on the fidelity and timing of spike transmission rather than on adaptivity or stateful dynamics. It retains biological interpretability through frequency filtering, realistic pulse width, and clean comparator-based spiking, while avoiding active memory or plasticity mechanisms. This makes it suitable for introductory neuromorphic studies, rapid prototyping, and modular extension.

5.3.4 Contribution and Uniqueness of the Proposed Design

The proposed model is distinct from prior neuromorphic implementations due to its specific emphasis on analog signal fidelity, transmission-stage realism, and biological interpretability. Unlike models that focus on spike timing variability, learning capability, or nanoscale memory, this work centres on a more fundamental question: whether biologically relevant signals can be filtered, thresholded, and transmitted through an analog neuron model without distortion or loss of information.

Previous models in literature often prioritise adaptability, material novelty, or dense integration. For instance, memristive neuron models such as those introduced by Yang et al. [3] or Hu and Liu [4] simulate complex threshold behaviours and dynamic attractors. However, these designs typically operate using synthetic input signals, and they do not offer stage-by-stage analysis of signal degradation. Their outputs are often assessed in terms of firing frequency or pattern types, rather than waveform quality or integrity across compartments.

In contrast, this project introduces a modular analog neuron that separates dendritic filtering, spike generation, and axonal transmission into clear and testable blocks. Each block is designed and evaluated using both time-domain and frequency-domain methods, providing a transparent view of how analog signals behave through the neuron pathway. The use of standard input signals such as Hodgkin-Huxley waveforms, EEG segments, and PQRST traces allows for direct comparison with biological behaviour and ensures that circuit performance is grounded in realistic physiological contexts.

A particularly novel aspect of this design is the validation of signal-to-noise ratio and spike shape at three separate circuit stages. This level of granularity is not commonly observed in prior neuron models. While many designs produce spikes in response to varying stimuli, few explicitly measure how much information is lost or preserved during the process. By quantifying SNR loss and pulse width changes from dendrite to soma to axon, this model confirms that its signal fidelity is both predictable and biologically acceptable.

Additionally, the proposed model avoids reliance on digital control, embedded code, or clocked sequencing. All circuit behaviour emerges naturally from analog component interactions. This enhances the model's value in low-power systems, analog signal processing modules, and education-focused hardware demonstrations. It can be used in simulation, bench-top testing, or embedded systems without requiring digital co-processors or calibration logic.

The modularity of the design further enhances its practicality. Each functional stage can be replaced, scaled, or extended independently. This makes the model easier to integrate into broader neuromorphic architectures or to adapt for research purposes. For example, the comparator threshold can be modified without affecting dendritic filtering, and the axonal amplifier can be tuned for different output loads without altering spike timing.

In summary, this model offers a unique contribution by focusing on signal transformation clarity, waveform preservation, and hardware simplicity. It fills a gap left by more complex neuron models that either obscure internal dynamics or focus heavily on learning behaviours. The combination of analog transparency, biological input compatibility, and waveform-focused evaluation makes it a strong candidate for future neuromorphic design work, particularly in contexts where clean signal handling and modular analysis are essential.

5.4 Future Improvements and Applications

The proposed analog neuron model has successfully demonstrated biologically inspired signal processing using discrete circuit components. Its ability to replicate key stages of neural behaviour shows promise for both simulation and real-world applications. Nonetheless, the model can be further developed to improve its fidelity, flexibility, and relevance across biological and engineering contexts. This section outlines recommended technical improvements and summarises the model's application potential.

5.4.1 Future Improvements

While the current implementation meets the primary goal of simulating biologically realistic signal transformation in hardware, several enhancements can improve its performance, adaptability, and biological accuracy:

- **Adaptive Thresholding Mechanisms:** Introducing threshold circuits that adjust based on recent input history would allow simulation of neuronal adaptation or fatigue. This could be achieved using integrator-based feedback or programmable references in the comparator stage.

- **Pulse Width and Shape Control:** The present comparator generates spikes of fixed width. Including monostable circuits or variable pulse-shaping units would enable emulation of diverse firing patterns and facilitate downstream spike encoding tasks.
- **Multi-Input Dendritic Integration:** Expanding the dendritic stage to receive and combine multiple analog inputs with different weights would support the modelling of excitatory and inhibitory synaptic integration, improving biological realism.
- **Tuneable Propagation Characteristics:** The axon stage may be extended with gain control, delay tuning, or regenerative buffering to simulate transmission properties over longer distances. This would allow exploration of axonal attenuation, conduction velocity, and signal regeneration.
- **Improved Signal Fidelity and Robustness:** Future iterations can include better PCB design, shielding, and component selection to minimise noise, voltage drop, and parasitic interference. This is especially important for scaling the model to neuron arrays or interfacing with sensitive analog inputs.

These improvements aim to enhance the biological relevance and system integration capability of the model without compromising its transparency and modularity.

5.4.2 Applications

The modular structure, analog operation, and compatibility with physiological signals position the proposed model as a versatile platform with near-term and long-term application potential. These applications can be grouped into three levels of maturity: immediate educational use, medium-term engineering integration, and longer-term biomedical prospects.

A. Short-Term Educational and Research Applications

The clarity of the dendrite–soma–axon separation makes this model highly suitable for laboratory demonstrations and teaching environments. Students can directly observe how analog circuits replicate core neural functions such as filtering, thresholding, and spike generation. The circuit also provides a transparent research platform for exploring how parameter changes affect spike dynamics, offering a useful testbed for neuromorphic circuit design.

B. Medium-Term Engineering and Neuromorphic Applications

As a low-power analog front-end, the model can serve as a spike encoder in wearable or embedded systems where continuous biosignals must be converted into discrete events for further processing. Its modularity supports benchmarking of alternative analog neuron designs under consistent conditions, enabling fair performance comparisons. The spike outputs can also be integrated into hybrid analog–digital neuromorphic systems, providing biologically interpretable input streams for large-scale event-driven computation.

C. Long-Term Biomedical Applications

While speculative, the model provides a foundation for potential biomedical uses once extended with adaptive thresholds and plasticity. Feeding pathological inputs such as epileptic EEG or arrhythmic cardiac signals could demonstrate how abnormal patterns alter spike behaviour, supporting disease modelling. With further refinement, the model could be adapted for prosthetic or implantable systems where hardware neurons assist or replace malfunctioning biological counterparts.

5.5 Technical Summary and Takeaways

This section summarises the key technical contributions and design insights from the project, highlighting how the proposed model achieves its stated objectives.

5.5.1 Summary of Design and Functionality

The model replicates the three core stages of a biological neuron using analog hardware: dendritic filtering, threshold-based soma spiking, and axonal pulse regeneration. The circuit responds predictably to real-world signals including sinusoidal, EEG, and cardiac inputs. Filtering behaviour, spike initiation, and pulse output were validated using Multisim simulations and MATLAB frequency analysis. Spike shape, timing, and signal-to-noise characteristics were preserved across all stages.

5.5.2 Design Takeaways and Engineering Insights

- **Modularity:** Each stage operates independently, supporting tuning, replacement, and future scaling.
- **Signal Transparency:** Analog signals are observable at every node, aiding validation and debugging.
- **Comparator Reliability:** The soma stage consistently produces low-jitter spikes across input types.
- **Pulse Integrity:** The axon stage ensures clean and repeatable digital-like pulses.
- **Adaptability:** The design supports both research experimentation and educational demonstration.

5.5.3 Alignment with Project Objectives

All design goals were achieved:

- Dendritic filtering was implemented using a tuned bandpass network.
- Soma behaviour was accurately simulated using a threshold comparator.
- Axonal regeneration was achieved through pulse-shaping circuitry.
- Biological interpretability was maintained throughout.
- The model offers practical pathways for extension and integration.

This summary confirms that the proposed model meets its biological and engineering objectives, offering a functional and modular analog framework for neuron transmission that is suitable for research, application, and teaching.

Chapter 6: Project Management and Development Summary

6.1 Timeline and Weekly Progress

The development of the proposed analog neuron model was executed over a structured 10-week period. Each week focused on a specific subset of tasks ranging from signal preprocessing and circuit prototyping to frequency analysis, spiking accuracy, and transmission continuity. This progressive workflow ensured that the design could evolve incrementally while still aligning with biological plausibility and analog feasibility.

Figure 5.6 presents the complete week-by-week development plan. It summarises the technical milestones achieved, identifies key risks, and outlines mitigation strategies that were employed throughout the design process.

Week	Date	Progress Summary	Risk	Mitigation
1	09/06 – 15/06	- Defined project objectives - Conducted literature review across all modelling goals - Planned weekly development milestones	–	–
2	16/06 – 22/06	- Created HH-based, ECG-based, and synthetic input signals - Converted signals to Multisim-compatible format using MATLAB - Drafted initial project brief	Input signals may not align with simulator input structure	Verified waveform compatibility in MATLAB before Multisim use
3	23/06 – 29/06	- Designed second-order Butterworth filter (Sallen-Key topology) - Tested across multiple inputs and RC combinations - Replicated axonal low-pass filtering	Filter behaviour may not generalise across signal types	Validated using EEG, HH, and ECG signals
4	30/06 – 06/07	- Introduced multiple input frequencies for biological realism - Performed FFT comparison between MATLAB and Multisim - Implemented dendritic coupling with multiple inputs - Discarded comparator-based soma due to low output swing	Spectral mismatches across platforms	Used MATLAB FFT to verify Multisim analysis
5	07/07 – 13/07	- Sampled frequencies from EEG signals - Introduced synaptic weights using potentiometers - Attempted multiplexing using MUX, SPST, and pulse gating - Determined multiplexing circuits were not biologically appropriate	Multiplexing produced non-biological effects	Removed multiplexing and documented reasoning
6	14/07 – 20/07	- Designed mixer circuit to isolate 10 Hz signals - Implemented MFB filter for sharper frequency response - Tuned RC values to reduce noise and distortion - Studied propagation through axon	Low-frequency selectivity was limited	Adopted MFB design for better filter control
7	21/07 – 27/07	- Upgraded axon design using comparator, transistor, and RC bridge - Explored diode-based non-linearity in dendrites (unsuccessful) - Focused on clean spike generation using analog components	Diodes failed to operate under low voltage inputs	Removed diodes and relied on active circuits
8	28/07 – 03/08	- Redesigned dendritic filters to stabilise behaviour post-axon - Achieved consistent soma spiking - Began transmission line modelling	Axonal feedback disrupted dendritic response	Isolated filter stages and tuned parameters
9	04/08 – 10/08	- Designed RC-based transmission line - Used Schmitt triggers and amplifiers to regenerate spikes (Node of Ranvier) - Observed digital spike propagation with low distortion	Spiking inconsistencies in early designs	Repeated simulations with tuned tolerances
10	11/08 – 17/08	- Finalised transmission model - Validated frequency preservation and spike fidelity - Completed graphical and mathematical checks - Model finalised for demonstration and submission	–	–

Table 5.6: Weekly Project Timeline with Identified Risks and Mitigation Strategies

During the early phases, significant effort was dedicated to sourcing and preparing biologically relevant inputs. MATLAB was used to generate and verify these signals before importing them into Multisim for circuit simulation.

By Weeks 3 to 6, several filter designs were explored and validated, including Butterworth and MFB (Multiple Feedback) configurations. Challenges such as filter misbehaviour, multiplexing failure, and diode-based distortion were encountered and resolved through critical design decisions and literature-driven modifications. Notably, biologically unrealistic configurations were actively discarded and documented.

From Week 7 onwards, the focus shifted to pulse generation and transmission. Schmitt triggers and RC configurations were used to emulate axonal spiking behaviour and signal regeneration, comparable to nodes of Ranvier. The final design demonstrated clean spike propagation with minimal distortion, satisfying the core objectives of the project.

6.2 Design Iterations and Learning Outcomes

Multiple iterations were performed throughout the project to test and refine key components. The initial dendritic filters were revisited following the addition of axonal circuits, leading to improvements in frequency selectivity and signal shaping. MUX-based multiplexing was excluded after proving inconsistent with natural synaptic behaviour.

These iterations provided important learning outcomes:

- **RC tuning** is critical in defining biologically relevant cutoffs and filtering behaviour.
- **Active components** such as Schmitt triggers provide stable and biologically consistent spike shaping.
- **Diode-based non-linearities**, while common in signal gating, are not well suited for low-voltage neural signals.
- **Transmission line modelling** requires fine-grained control over impedance and delay characteristics to preserve waveform shape.

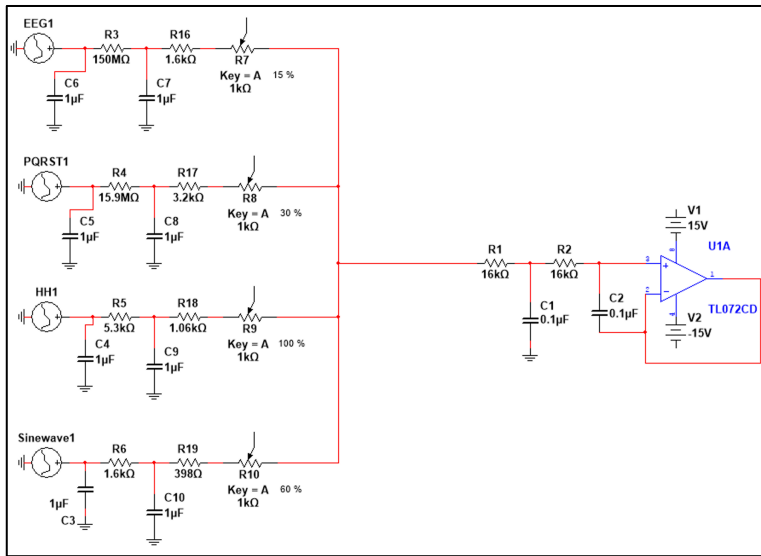
These insights strengthened the final model's fidelity, not only in terms of performance but also in its alignment with biological neuron transmission processes.

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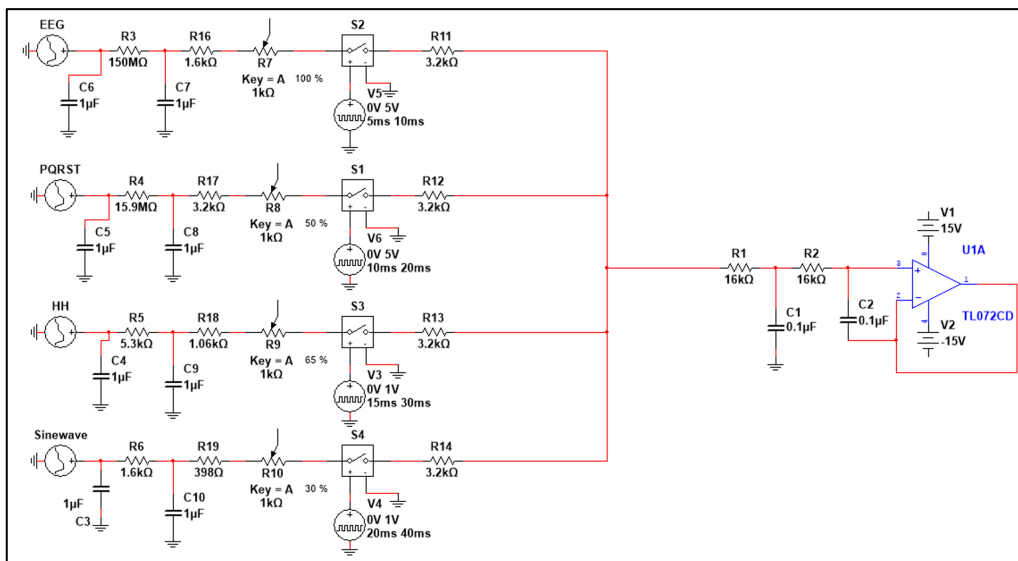
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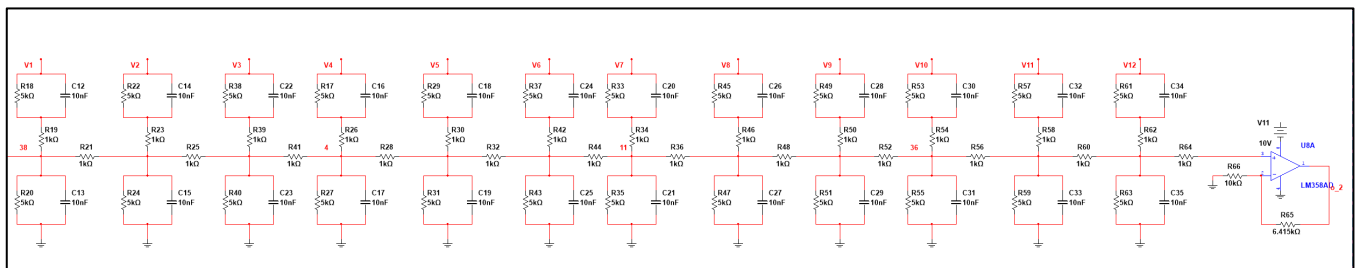
Appendixes



Design testing with potentiometers



Design with SPST switches



Model to find the position of amplifiers (Node of Ranvier)